Fuzzy PI Control for bittide: Nonlinear Analysis and Transient Performance

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Abstract

Finite receiving buffers make bittide synchronization safety depend on initial clock frequency differences and available storage reserve. We propose a decentralized incremental Fuzzy PI controller and derive configuration-dependent sufficient conditions connecting mismatch, buffer targets, graph structure, timing and controller scales. An explicit PI calibration and exact nonlinear map support a controlled comparison. For a synchronous zero-delay fluid reference, Lyapunov analysis and outward interval prefix/tail bounds establish a strictly larger interval of compatible safe occupancy targets for the prescribed initialization. Full parameter-cell coverage certifies Fuzzy excursion below 17.342 frames throughout a fixed-direction 399.6–400.4-ppm interval, while PI's attained peak exceeds 23.380 frames. At target 20, Fuzzy is therefore strictly safe and unprotected PI crosses zero. Theory-directed reference simulations agree with these predictions. Twelve continuous-phase Callisto runs separately assess delayed integer execution: at target 83 and 1600-ppm spread, Fuzzy remains in $[11, 155]$ without intervention, while PI reaches zero and requires five guard episodes. Upper-target cases expose an additional delayed-credit constraint. These results establish a conditional safety advantage and distinguish reference guarantees from finite implementation evidence.

CCS Concepts

• **Computing methodologies**;

Keywords

Distributed synchronization, bittide, fuzzy control, logical synchrony, cyber-physical systems

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1 Introduction

bittide is a hardware synchronization mechanism that realizes logical synchrony in distributed systems using independent local physical clocks and continuous frame exchange [3]. For applications that depend on reproducible execution, the relationships between communicated events must remain well defined as physical timing varies. The implementation challenge in bittide is to maintain these logical relationships while accommodating clock frequency differences, link latency and finite buffer capacity.

In a bittide network, elastic receiving buffers connect the physical progress of neighboring nodes to their logical execution without requiring a global physical clock. Each node adjusts its local clock frequency using locally observed buffer occupancies. The buffers therefore serve two coupled roles: they provide feedback for frequency synchronization and absorb the imbalance between incoming frames and local consumption. Clock frequency differences and changes in frame-delivery time affect the same finite storage that supports continued logical execution. The controller must consequently be evaluated together with the buffer trajectories it produces [3].

The resulting bittide control problem concerns the entire trajectory towards an operating point. Initial clock frequency differences consume buffer headroom before the network settles. Small steady-state frequency differences alone do not establish that buffer occupancy remains within capacity throughout the transient. The design question is how initial mismatch, available buffer reserve and controller configuration jointly determine safe operation, and whether a nonlinear local controller can admit safe configurations that its calibrated PI counterpart cannot. This requires relating controller dynamics to capacity constraints, while retaining positive physical frequencies and respecting correction limits.

Existing bittide research establishes important foundations for this problem. PI performance analysis [1] relates closed-loop behavior to controller gains and network resistance distances. Fluid occupancy analysis [4] characterizes steady buffer levels, and reframing methods [2] address buffer centering. Our earlier study examined PI gain selection under topology and fixed-delay constraints. Building on these results, we study configuration-dependent buffer safety for a proposed nonlinear Fuzzy PI law. Its rule surface must be connected to a closed-loop safety argument and an explicit PI comparison: a larger instantaneous correction or a smaller simulated peak alone does not establish a larger admissible operating domain.

We propose a decentralized occupancy-based Fuzzy PI controller for bittide. Each node computes a frequency correction increment from its receiving-buffer error and the update-to-update change in that error. In the central input region studied here, the proposed surface strengthens its calibrated linear counterpart for same-sign inputs and attenuates its magnitude for opposing-sign inputs. This direction-dependent action provides flexibility at fixed calibrated PI gain coordinates. It motivates investigating whether the nonlinear correction can reduce transient buffer demands while retaining convergence, and determining how its effects on inter-node frequency differences and control activity depend on the operating conditions.

We first derive a joint reference-model safety test linking initial mismatch energy to buffer reserve, graph structure, controller scales and timing. For synchronous zero-delay fluid execution, an

invariant input region and common Lyapunov certificate establish convergence, positive frequencies and correction bounds. Outward interval propagation and a contracting infinite tail then prove a strict excursion benefit at matched gains. At fixed capacity, a compatible-target translation lemma converts this benefit into a strictly larger safe target interval for the prescribed initialization. Complete parameter-cell coverage additionally establishes safety separation throughout a fixed-direction 399.6–400.4-ppm interval: Fuzzy excursion is below 17.342 frames, while PI attains more than 23.380 frames. At a common initial/target occupancy of 20, Fuzzy remains strictly safe and unprotected PI crosses zero.

The simulations are designed to check these deductions. Thirty-four new reference trajectories and six verified cached sources check the original control rules, occupancy translation, mismatch boundaries and distribution-dependent energy. Twelve continuous-phase Callisto runs separately restore local-clock polling, delayed actuation and integer frame delivery. At 1600-ppm initial spread and target 83, Fuzzy retains occupancy in [11, 155] without a guard, while PI reaches zero and invokes five input-empty guard episodes. Upper-target runs expose a delayed outgoing-credit constraint even before receiving buffers fill. These qualitative results support the conditional margin mechanism and identify the execution assumptions needed for stronger guarantees. Earlier independently tuned disturbance comparisons are retained as supplementary evidence.

2 Background

2.1 The bittide mechanism and logical synchrony

bittide realizes logical synchrony through ordered frame exchange between nodes driven by independent local physical clocks. Logical synchrony specifies relationships between distributed events without requiring equal physical phases or a global physical clock [3]. During normal unstalled operation, nodes send frames to their neighbors and consume frames from their receiving buffers as their local clocks advance. Elastic receiving buffers accommodate the difference between frame delivery and local consumption while preserving frame order.

The bittide controller uses these receiving-buffer occupancies as local feedback to adjust clock frequency. Faster incoming progress tends to fill a buffer, whereas faster local consumption tends to drain it; link-latency changes also alter frame delivery. Frequency control therefore regulates physical progress while buffer availability supports the prescribed logical execution. Frequency agreement alone does not prove the required frame relationships or application-payload correctness.

2.2 Continuous clocks and buffer dynamics in bittide

Each node maintains a local physical clock driven by a frequency-controlled oscillator. In our bittide model, node i has continuous clock phase $\theta_i(t)$ and physical frequency $\omega_i(t) = \omega_i^u + c_i(t)$, where ω_i^u is its constant uncontrolled (uncorrected) clock frequency and c_i is the applied frequency correction, or control input. In Callisto simulations, each node holds corrections between local actuation events; phase is continuous and its slope changes at those events. Integer crossings generate frames inside this continuous plant. Positive physical operation requires $\omega_i > 0$; a negative correction is not a negative frequency.

For edge $j \to i$, occupancy β_{ji}, initial occupancy $\beta_{ji}(0)$, arrival count A_{ji}^{arr} and consumption count N_i obey, while progress guards remain open,

$$\begin{aligned} \beta_{ji}(t) &= \beta_{ji}(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \\ N_i(t) &= \lfloor \theta_i(t) \rfloor - \lfloor \theta_i(0) \rfloor. \end{aligned} \tag{1}$$

Arrivals include frames already in flight at initialization. Fluid models replace integer counts by phase increments; at constant physical link latency τ_{ji} they give $\dot{\widetilde{\beta}}_{ji} = \omega_j(t - \tau_{ji}) - \omega_i(t)$. This analytical approximation is distinguished from actual integer execution [1, 4].

2.3 Safety, syntony and buffer centering

For each receiving edge e, let $\beta_{\max,e}$ denote its buffer capacity; a common capacity is written $\beta_{\max}$. With correction limit $c_{\max}$, normal operation seeks $0 < \beta_e(t) < \beta_{\max,e}$, $\omega_i(t) > 0$ and $|c_i(t)| \le c_{\max}$ along the trajectory. A protection-induced stall is distinct from successful unstalled control. In the nominal reference, frequency synchronization (syntony) is expressed by $\Delta_\omega = \max_i \omega_i - \min_i \omega_i \to 0$, together with occupancy regulation, $\beta_e \to \beta_e^*$, for compatible targets. At a half-capacity target this is buffer centering; the safety analysis also varies compatible initial/target occupancy at fixed capacity. These objectives do not mean phase equality or tracking frequency one. Under persistent delivery disturbances, we assess residual frequency differences and buffer reserves rather than exact equilibrium. Buffer headroom denotes the remaining reserve before underflow or overflow.

Existing PI analysis relates performance to gains and resistance distances [1]; occupancy and reframing analyses characterize limiting levels and centering [2, 4]. For bittide, the additional question is how a fully specified nonlinear local law changes transient buffer use and disturbance response at controlled scales while respecting the execution constraints above. Appendix A expands these distinctions.

3 Proposed Fuzzy PI Controller

3.1 Local measurements and continuous-clock interface

Let $\mathcal{N}_i^-$ be the fixed set of incoming neighbors and $d_i = |\mathcal{N}_i^-| > 0$. For common capacity $\beta_{\max}$ and target β_{ji}^*, define frame-sum and percentage errors

$$E_{i,k} = \sum_{j \in \mathcal{N}_i^-} (\beta_{ji,k} - \beta_{ji}^*), \quad e_{i,k} = -\gamma_i E_{i,k}, \quad \gamma_i = \frac{100}{d_i \beta_{\max}}. \tag{2}$$

Thus positive E calls for faster local consumption. The index k is local to each node; e is target minus measured average occupancy in percentage points. Unequal capacities require consistent weighted errors, rather than the scalar calibration below.

With local polling period p and execution delay d_{ctrl}, measurement and application times satisfy

$$\begin{aligned} \theta_i(t_{i,k}^m) &= \theta_i(0) + kp, \\ \theta_i(t_{i,k}^a) &= \theta_i(0) + kp + d_{\mathrm{ctrl}}. \end{aligned} \tag{3}$$

The product kp is update index times period, not gain k_p. Requested correction $q_{i,k}$ is limited to $c_{i,k} = \mathrm{sat}_{c_{\max}}(q_{i,k})$ and held until the next application, with $\dot{\theta}_i = \omega_i^u + c_{i,k}$. The physical measurement-to-actuation age therefore depends on the past clock frequency. Initial correction is zero. Requested and applied corrections coincide when the limiter is inactive: $c_{i,k} = q_{i,k}$.

Actual arrivals use compatible negative-time phase history and an order-preserving delivery map. For desired send-time latency $\tau_{ji}(s)$, arrival time is $a_{ji}(s) = \sup_{r \le s}\{r + \tau_{ji}(r)\}$; counts follow delivered integer send crossings, not an arbitrary continuous cutoff. This permits ordered batch release and imposes no finite serialization rate. Appendix B specifies count initialization, the in-flight offset and the complete execution interface.

The runtime's existing guards are identical for PI and Fuzzy. A guard pauses data phase ψ_i, while physical phase θ_i and controller updates continue; wholly unstalled execution has $\psi_i = \theta_i$. For constant forward and reverse-feedback latencies, the used outgoing credit χ_{ji} satisfies

$$\chi_{ji}(t) = \beta_{ji}(t) + n_{ji}^{\mathrm{flight}}(t) + n_{ji}^{\mathrm{unack}}(t). \tag{4}$$

Here n_{ji}^{flight} counts frames still in forward transit, and n_{ji}^{unack} counts consumed frames not yet reflected in the sender's reverse feedback. A further send requires $\chi_{ji} + 1 \le \beta_{\max}$. Hence receiving occupancy alone does not establish guard-free execution. Appendix D derives this identity for the constant-latency validation model.

3.2 Fuzzification, inference and control update

The implemented integrate-first PI baseline uses sample sum I, fixed integration scale $h = p/\omega_{\mathrm{base}}$, and $\kappa_I = k_i h$:

$$I_k = I_{k-1} + E_k, \qquad q_k^{\mathrm{PI}} = k_p E_k + \kappa_I I_k. \tag{5}$$

The sum has no elapsed-time multiplier. The proposed Fuzzy law is

$$\begin{aligned} X_k &= K_e e_k, \quad Y_k = K_{\Delta e}(e_k - e_{k-1}), \\ A_k &= A_{k-1} + K_u F(X_k, Y_k), \qquad q_k^{\mathrm{F}} = A_k, \end{aligned} \tag{6}$$

where $K_e, K_{\Delta e} > 0$, $K_u < 0$ and A stores the requested correction. The second input is a sample difference, not a time derivative. All rules and gains remain fixed during a run.

Fuzzification uses five labels: negative large (NL), negative small (NS), zero (ZE), positive small (PS) and positive large (PL). Their input centers ζ_a are $(-10, -5, 0, 5, 10)$. Interior memberships are triangular: $\varphi_a(z) = [1 - |z - \zeta_a|/5]_+$, $a = 2, 3, 4$, where $[z]_+ = \max(z, 0)$. The outer shoulders are

$$\varphi_1(z) = \mathrm{clip}_{[0,1]}\,\frac{-z-5}{5}, \quad \varphi_5(z) = \mathrm{clip}_{[0,1]}\,\frac{z-5}{5}. \tag{7}$$

Here $\mathrm{clip}_{[0,1]}$ projects onto $[0, 1]$. Consequent singleton values are $s = (-5, -2.5, 0, 2.5, 5)$. Inference uses Mamdani minimum rule activation and singleton weighted-average defuzzification. All 25 rules are specified by $\ell(a, b) = \min(5, \max(1, a + b - 3))$ and

$$w_{ab} = \min(\varphi_a(X), \varphi_b(Y)), \quad F = \frac{\sum_{a,b=1}^{5} w_{ab} s_{\ell(a,b)}}{\sum_{a,b=1}^{5} w_{ab}}. \tag{8}$$

Rules sharing a consequent are summed separately; replacing their sum by maximum aggregation changes the controller. The denominator is positive for finite inputs, and $|F| \le 5$ bounds each increment, not the indefinitely accumulated correction. Initialize $I_{-1} = A_{-1} = e_{-1} = 0$. External clipping retains raw A; anti-windup, observer and mode switching are disabled. Table 1 also gives the complete consequent matrix. This specification fixes the implemented nonlinear map; fuzzy inference is not used here as an unspecified online gain tuner.

Table 1: All consequent labels of the proposed Fuzzy law. Rows and columns are the present-error and error-difference inputs, respectively.

$X\backslash Y$	NL	NS	ZE	PS	PL
NL	NL	NL	NL	NS	ZE
NS	NL	NL	NS	ZE	PS
ZE	NL	NS	ZE	PS	PL
PS	NS	ZE	PS	PL	PL
PL	ZE	PS	PL	PL	PL

3.3 PI calibration and nonlinear mechanism

For $F_0(X, Y) = (X+Y)/2$, coefficient matching with $\Delta q_k^{\mathrm{PI}} = k_p \Delta E_k + \kappa_I E_k$ gives

$$k_p = -\frac{K_u \gamma_i K_{\Delta e}}{2}, \qquad \kappa_I = -\frac{K_u \gamma_i K_e}{2}. \tag{9}$$

Proposition 3.1 (Calibrated linear counterpart). *With* (9)*, common measurements, consistent zero constructors and identical execution policies, replacing F by F_0 gives the PI baseline's requested correction at every update, including $q_0 = (k_p + \kappa_I)E_0$.*

This is a scale and implementation equivalence for F_0, not full nonlinear equivalence. The substitution and initialization proof are in Appendix B.

This mapping controls a common comparison pitfall: matching gain names without matching the measured error, integration order and time scale does not match controllers. For the triangle, $\gamma = 100/4096$ and $h = 1000/(29999/30000)$, so the moderate coordinates $(k_p, k_i) = (2 \times 10^{-6}, 5 \times 10^{-11})$ correspond to $K_e = 0.1$, $K_{\Delta e} \simeq 3.9998667$ and $K_u \simeq -4.0961365 \times 10^{-5}$. Node indices were suppressed in the update equations. On unequal-degree graphs, node-dependent $K_{u,i}$ can compensate γ_i when common calibrated coordinates are required. Target and capacity conventions must stay consistent.

Proposition 3.2 (Central nonlinear map). *On $\Omega_r = \{(X, Y) : |X| + |Y| \le r\}$, $0 < r \le 5$, set $m = \min(|X|, |Y|)$ and*

$$J = \begin{cases} \mathrm{sgn}(X)m, & XY > 0, \\ 0, & XY \le 0. \end{cases} \qquad F(X, Y) = \frac{F_0(X, Y) + J}{1 + 2m/5}. \tag{10}$$

There exists $\alpha(X, Y)$ with $F = \alpha F_0$ and $\underline{\alpha} = (1 + r/5)^{-1} \le \alpha \le 2 = \overline{\alpha}$.

The four active min-rule weights yield this identity. Same-sign inputs strengthen F_0; opposing signs attenuate its magnitude. Uniformly increasing PI gains cannot produce both effects. This axis calibration is not an ordinary Jacobian linearization; details and sector bounds are in Appendix B.

With $\mathcal{R}_F = F - F_0$, the zero-constructor raw correction obeys

$$A_k = k_p E_k + \kappa_I \sum_{\ell=0}^{k} E_\ell + K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{11}$$

The last term exposes nonlinear memory. It is accounting on Fuzzy's own measurements, not a causal decomposition of two independently evolving closed loops. Moreover, same-sign inputs do not always mean growing absolute error. Performance requires closed-loop evidence.

4 Configuration-Dependent Buffer Safety

4.1 Declared reference and exact recurrence

The analysis uses a connected, undirected, unit-weight graph with Laplacian L, common capacity $\beta_{\max}$, zero physical link latency, zero control-execution delay, fluid occupancies and synchronous physical sampling $t_k = kH$. Corrections are zero-order held. Initial phases are equal, and targets satisfy $\beta^*_{ji} = \lambda_{ji}$ for compatible additive markings. These are reference assumptions, not properties of actual asynchronous Callisto. The endpoint recurrence is exact for continuous held flow, not Euler integration or a discrete firing-count plant.

Set $\delta\omega = \omega^u - \omega_{\mathrm{ref}}\mathbf{1}$, $x_k = \theta(t_k) - \theta(0) - kH\omega_{\mathrm{ref}}\mathbf{1}$, $w_k = \delta\omega + c_k$ and $v_k = Hw_{k-1}$ for $k \geq 1$. Then $E_k = -Lx_k$ and $\beta_{ji}(t_k) - \beta^*_{ji} = x_{j,k} - x_{i,k}$. With $D_{\alpha,k} = \mathrm{diag}(\alpha_{i,k})$, the calibrated law yields

$$\begin{aligned} v_{k+1} &= v_k - D_{\alpha,k}(bLx_k + aLv_k), \\ x_{k+1} &= x_k + v_{k+1}, \quad a = Hk_p, \quad b = Hk_ih. \end{aligned} \tag{12}$$

The integration scale h and physical period H remain distinct. Zero controller memory and centered startup give $c_0 = 0$ and the genuine bootstrap $x_1 = v_1 = H\delta\omega$; stored previous error is not a previous plant interval. Appendix C derives these relations.

4.2 Invariant domain and safety certificate

Write $L = U\Lambda U^\top$ with $U^\top\mathbf{1} = 0$ and $U^\top U = I$. The positive diagonal matrix Λ contains the nonzero graph eigenvalues $\mu_2, \ldots, \mu_n$. Define $\xi = \Lambda^{1/2}U^\top x$, $\eta = \Lambda^{1/2}U^\top v$ and $y = (\xi^\top, \eta^\top)^\top$. The state-dependent symmetric matrix $M_k = \Lambda^{1/2}U^\top D_{\alpha,k}U\Lambda^{1/2}$ has eigenvalues in $[\nu_-, \nu_+] = [\underline{\alpha}\mu_2, \overline{\alpha}\mu_n]$. No commutation between D_α and L is assumed. Define

$$B(\nu) = \begin{bmatrix} 1 - b\nu & 1 - a\nu \\ -b\nu & 1 - a\nu \end{bmatrix}, \quad V_k = y_k^\top(\bar{P} \otimes I)y_k. \tag{13}$$

Here $\bar{P} \succ 0$ is a 2×2 common Lyapunov witness and $\otimes$ denotes the Kronecker product.

Lemma 4.1 (Endpoint contraction). *If $0 < \rho < 1$ and*

$$B(\nu_\pm)^\top \bar{P} B(\nu_\pm) \preceq \rho\bar{P}, \tag{14}$$

then every update in the regional sector satisfies $V_{k+1} \leq \rho V_k$, including time- and node-dependent factors.

Affine dependence of B gives matrix convexity over the whole endpoint interval. Orthogonal diagonalization of each symmetric M_k preserves $\bar{P} \otimes I$, so one witness covers changing node factors. Appendix C supplies the full proof.

The importance of a common witness is that the nonlinear surface changes its factors at every update. Stability of one frozen-gain characteristic polynomial would not certify the resulting trajectory. The symmetric projection handles the node dependence without assuming that all factors or graph modes evolve independently. However, this certificate remains conditional on the inputs staying in their derived sector; the following radius test closes that condition.

For $\gamma_i = 100/(d_i\beta_{\max})$, define

$$\Psi_i = \gamma_i^2 d_i \max_{\sigma=\pm1} [K_e, \sigma K_{\Delta e}]\bar{P}^{-1}[K_e, \sigma K_{\Delta e}]^\top. \tag{15}$$

If $V_1 \leq R$ and $R\Psi_i \leq r^2$ for every node, the energy output bound gives $|X_i| + |Y_i| \leq r$. Together with contraction, induction proves input-domain invariance rather than assuming it. Let $L^\dagger$ be the Laplacian pseudoinverse and $R^{\mathrm{eff}}_{ij} = (\mathbf{e}_i - \mathbf{e}_j)^\top L^\dagger(\mathbf{e}_i - \mathbf{e}_j)$ the effective resistance. Put $R^{\mathrm{eff}}_{\max} = \max_{i \neq j} R^{\mathrm{eff}}_{ij}$ and $m_e = \min(\beta^*_e, \beta_{\max} - \beta^*_e)$. Then

$$\begin{aligned} |\beta_e - \beta^*_e|^2 &\leq R[\bar{P}^{-1}]_{11}R^{\mathrm{eff}}_e, \\ \Delta_\omega(t) &\leq D(R)\rho^{k/2}, \quad t \in [t_k, t_{k+1}), \\ D(R) &= H^{-1}\sqrt{R[\bar{P}^{-1}]_{22}R^{\mathrm{eff}}_{\max}}. \end{aligned} \tag{16}$$

Here R^{eff}_e uses the endpoints of edge e. Affine intersample occupancy covers all physical times.

The unprojected common frequency also needs a bound. Set $g = (\kappa_I, k_p)^\top$, $s_\alpha = (\overline{\alpha} - \underline{\alpha})/2$, $\bar{w}_0 = n^{-1}\mathbf{1}^\top\delta\omega$ and

$$\begin{aligned} W_c &= |\bar{w}_0| + \frac{2s_\alpha}{1-\rho}\sqrt{\frac{\mu_n R}{n} g^\top \bar{P}^{-1} g}, \\ W_d &= H^{-1} \max_i \sqrt{R[\bar{P}^{-1}]_{22}(L^\dagger)_{ii}}. \end{aligned} \tag{17}$$

The following joint sufficient conditions close capacity, limiter and positive-frequency requirements:

$$\begin{aligned} &V_1 \leq R, \quad R\Psi_i \leq r^2, \quad 0 < \beta^*_e < \beta_{\max}, \\ &R[\bar{P}^{-1}]_{11}R^{\mathrm{eff}}_e < m_e^2, \\ &W_c + W_d + \|\delta\omega\|_\infty < c_{\max}, \\ &\omega_{\mathrm{ref}} - W_c - W_d > 0. \end{aligned} \tag{18}$$

Corollary 4.2 (Reference safety and convergence). *Under the reference assumptions, endpoint certificate and joint conditions, all buffers remain strictly within capacity, the limiter is inactive and physical frequencies stay positive. Buffer deviations and pairwise frequency differences converge to zero. If a separate tolerance Δ_{safe} is specified, $D(R) < \Delta_{\mathrm{safe}}$ also certifies its all-time bound.*

Summable common-frequency increments give a finite limiting rate, which need not equal the mean uncontrolled frequency for Fuzzy. A failed sufficient test is inconclusive; these conditions are not necessary stability boundaries or a certificate for delayed integer execution.

4.3 How the parameter constraints interact

The certificate exposes relationships between initial mismatch, buffer reserve, graph structure and controller scales, rather than independent recommended ranges for each parameter. Define $c_P = [1,1]\bar{P}[1,1]^\top$. At the centered bootstrap,

$$V_1 = H^2 c_P\, \delta\omega^\top L \delta\omega. \tag{19}$$

Thus physical observation time and the mismatch direction enter the same initial-energy test. A statement about one maximum

frequency difference alone does not identify this energy on an arbitrary graph. In the worked triangle $c_P = 1$, but it need not equal one for another witness. The joint bounds must then accommodate both this initial energy and the nonlinear input region.

For example, increasing capacity at unchanged Fuzzy gains changes γ_i and therefore the frame-count input scaling. Changing the reference sampling interval changes a and b; reconfiguring local polling can additionally change h. These distinct effects must not be replaced by a change to one nominal gain alone. Adding neighbors changes the measured sum, normalization and graph spectrum together. Table 2 explains why consistent parameter changes require recertification, even when one individual inequality appears less restrictive.

The initial occupancy assumption is equally important. This theorem starts at compatible targets with zero controller memory. An arbitrary non-centered start introduces an affine error or different phase coordinates and can alter the first controller call. It cannot be certified by inserting a new occupancy value into the same centered bootstrap. Likewise, finite link latency/control-execution delay and asynchronous polling belong to the Callisto simulations here, not to a hidden delay margin in the zero-latency proof. These distinctions allow the mathematical conditions to be used as reproducible design tests without presenting them as absolute hardware limits.

4.4 Strict headroom comparison

For the triangle use $\beta_{\max} = 2048$, compatible centered marking 1024, $\omega^u = (1.0002, 0.9998, 0.9999)^\top$, $\omega_{\mathrm{ref}} = 29999/30000$, $H = h = 30000000/29999$ and $(k_p, k_i) = (2 \times 10^{-6}, 5 \times 10^{-11})$. The matched Fuzzy gains are $(K_e, K_{\Delta e}, K_u) = (0.1, 29999/7500, -768/18749375)$. A witness is

$$\bar{P} = \frac{1}{50000}\begin{bmatrix} 9 & 181 \\ 181 & 49629 \end{bmatrix}, \quad \rho = 1599/1600, \quad r = 1. \tag{20}$$

Exact rational endpoint checks and outward enclosures verify the joint conditions. A coarse excursion bound is 32.237442 frames; sharper interval propagation and a contracting infinite tail give:

Theorem 4.3 (Prescribed-initialization headroom benefit). *For this reference and zero controller constructors, both methods satisfy the safety and convergence conclusions above. Their all-time excursions $B_J = \sup_{t\ge 0,e} |\beta_e^J(t) - 1024|$ obey*

$$B_{\mathrm{F}} < 17.315 < 23.40370130812701 \le B_{\mathrm{PI}}. \tag{21}$$

Minimum two-sided Fuzzy headroom exceeds 1006.685 frames and improves on PI by more than 6.08870130812701 frames.

Proof outline. PI reduces to an exact rational scalar recurrence with an attained prefix peak at sample 111. Outward interval propagation encloses the full nonlinear Fuzzy prefix through sample 200. Tail bounds of 17.958057 frames (PI) and 12.292711 frames (Fuzzy) are below their respective prefix peaks. Affine intersample flow excludes a missed larger peak. Appendix C contains the rational witness, interval arithmetic and complete proof. □

What closes the infinite horizon. The initial energy is the exact fraction $V_1 = 234000000/899940001$. With this radius the analyzed inputs satisfy $|X| + |Y| < 0.171455 < 1$, requested correction remains below 0.004166148 and physical frequency stays above 0.99603385. These bounds establish that the unconstrained trajectory never reaches the limiter, closing an assumption that otherwise could be circular. The common-frequency bound is also necessary: a network of equally nonpositive rates would have zero frequency spread but invalid progress.

The strict ordering compares Fuzzy's certified upper bound with PI's attained prefix peak. Both infinite-tail bounds are below their prefix peaks, and intersample occupancy is affine. Comparing only conservative upper bounds would not establish this ordering; Appendix C provides the complete arithmetic.

The excursion theorem fixes one initial frequency vector and matched gains. The following argument establishes a safety advantage for a family of configurations at fixed capacity and normalization. It does not compare independently optimized controllers.

4.5 Initial mismatch and the admissible occupancy interval

The joint conditions define a sufficient domain of configurations. For the centered bootstrap, let $Q = \delta\omega^\top L \delta\omega$ and $S_x = [\bar{P}^{-1}]_{11}$. Since each undirected connection is counted once in L,

$$Q = \sum_{\{i,j\}\in\mathcal{E}_u} (\omega_i^u - \omega_j^u)^2, \qquad V_1 = H^2 c_P Q. \tag{22}$$

Here $\mathcal{E}_u$ is the undirected edge set. Choosing $R = V_1$ makes the buffer and input-region tests explicit:

$$H^2 c_P Q S_x R_e^{\mathrm{eff}} < m_e^2, \qquad H^2 c_P Q \Psi_i \le r^2. \tag{23}$$

These tests require the endpoint, correction and positive-frequency conditions above. They show that initial mismatch and available storage are coupled; neither a frequency range nor a buffer capacity alone specifies a safe configuration. In particular, equal maximum initial frequency spreads can have different Q.

A sharper configuration comparison follows from the certified excursions. Use z for a common initial occupancy and target, distinguishing it from the recurrence coefficient $b = Hk_ih$. Keep capacity, gains, frequency vector, initial phases and timing fixed, and set all compatible zero-delay markings to z.

Corollary 4.4 (A strict advantage in admissible occupancy). *For the prescribed reference comparison, changing the common initial occupancy, target and compatible marking from 1024 to z leaves each controller's phase and frequency trajectories unchanged. The interval of targets with uniformly positive two-sided headroom is*

$$\mathcal{I}_J = (B_J, \beta_{\max} - B_J), \quad J \in \{\mathrm{PI}, \mathrm{F}\}. \tag{24}$$

Consequently $\mathcal{I}_{\mathrm{PI}} \subsetneq \mathcal{I}_{\mathrm{F}}$. At $\beta_{\max} = 2048$ and $z = 20$, Fuzzy has all-time occupancy in $(2.685, 37.315)$, whereas the unprotected PI reference crosses the lower capacity boundary.

Proof. In the unprotected fluid reference, $\beta_{ji}^{J,z} = z + x_j^J - x_i^J$. Subtracting target z cancels the shift from the measured errors. Fixed $\beta_{\max}$ preserves γ_i, so induction over controller updates preserves the inputs, memory and phase trajectories. Reverse channels have opposite deviations; the minimum two-sided headroom is therefore $\min(z, \beta_{\max} - z) - B_J$. The strict excursion ordering proves the interval inclusion and the stated safe/crossing case. The precision of the prefix/tail certificate, rather than the coarse Lyapunov buffer bound, establishes the Fuzzy capacity conclusion here. □

Table 2: Coupled roles of system parameters in the declared reference certificate. Every row must be evaluated for the same controller, graph and initialization; the table is not a set of independent necessary stability limits.

Parameter group	Role in the sufficient conditions
Initial clock frequency mismatch	$H^2 c_P \delta\omega^\top L \delta\omega \le R$ sets the required radius.
Initial occupancy and capacity	Compatible target $0 < \beta_e^* < \beta_{\max}$ and reserve m_e must contain the output bound.
Neighbour count and normalization	$\gamma_i = 100/(d_i \beta_{\max})$ enters Ψ_i, together with graph degree and Fuzzy input scales.
Topology	μ_2, μ_n determine spectral endpoints; effective resistance determines edge and pair outputs.
Control timing and gains	$a = Hk_p$, $b = Hk_i h$ enter the common endpoint inequalities; changing H changes the certificate.
Correction and absolute frequency	$W_c + W_d + \|\delta\omega\|_\infty < c_{\max}$ and $\omega_{\text{ref}} > W_c + W_d$ close execution constraints.
Marking and path history	Targets must match initialization; delayed in-flight frames cannot be silently removed when forming the zero-latency reference.

Table 3: Certified prefix and tail roles in the strict reference comparison. All entries are fluid frames; prefix enclosures are outward-rounded here.

Controller	Prefix peak enclosure	Tail upper bound
PI	[23.40370130812701, 23.40370130812702]	17.95805698665804
Fuzzy	[17.314056817962, 17.314056817963]	12.29271005372874

The same example connects the margin benefit to tolerable initial mismatch. Along $\delta\omega(s) = s(7, -5, -2)^\top/30000$, the initial spread is 400s ppm. In the non-clipped PI reference, linearity gives $B_{\text{PI}}(s) = sB_{\text{PI}}(1)$. For $z = 20$ its capacity threshold is approximately 341.8263 ppm, while Fuzzy is certified safe at the 400-ppm point. This establishes an advantage along this fixed mismatch direction. It does not certify every 400-ppm vector or a uniform Fuzzy bound for all smaller s. Appendix C.11 gives the full argument and parameter-domain interpretation.

The advantage also holds on a quantified parameter interval. Keeping the mismatch direction fixed, outward interval propagation on all 20 cells covering $s \in [0.999, 1.001]$ gives a uniform Fuzzy all-time excursion below 17.342 frames. PI's attained peak is above 23.3802976 frames throughout this interval. Thus the Fuzzy reference is safe at $z = 20$ for every initial spread from 399.6 to 400.4 ppm, while each corresponding unprotected PI trajectory crosses zero. This certifies a region in the (s, z) configuration slice, rather than inferring an interval from a finite grid of successful runs.

For an integer-valued compatible prefill z in this fluid reference, the bounds have a direct interpretation. Over the proved parameter interval, the minimum common prefill with positive uniform reserve is 18 for Fuzzy and 24 for PI. Fuzzy's bound is below 18, but 17 fails at the included $s = 1$ point; PI's exact scaled peak lies strictly between 23 and 24 throughout the interval. This concerns initial/target occupancy at fixed capacity, not a reduction in the configured physical buffer capacity.

5 Theory-Directed Simulation and Analysis

5.1 Validation design and execution models

The experiments check predictions derived before each batch, rather than tune gains to obtain a favorable comparison. We distinguish the proof-matched continuous fluid reference from continuous-phase Callisto. The reference uses the triangle, capacity $\beta_{\max} = 2048$, gains and physical period of Section 4, zero delays, zero memory and compatible initial occupancy/target/marking z. Fuzzy executes the original 25-rule implementation; its central formula is used in the separate proof, not as the sole simulator. The horizon is $T = 2001H$, comprising the bootstrap and 2000 further held-flow intervals. Full trajectories include every endpoint; affine interpolation covers intersample extrema.

Validation comprises 34 new reference trajectories, six verified cached sources and 12 new Callisto runs. The reference cases check target translation, mismatch boundaries, equal-spread directions and uniform-interval endpoints. These are qualitative consistency checks informed by theory and earlier results. Proof computations and translated maps are separate from plant runs; full protocols and counts appear in Appendix D.

Callisto keeps continuous physical phase, integer frame events, local polling $p = 1000$, execution delay $d_{\text{ctrl}} = 10$ and constant physical link latency 20. Initial phases are 0.1 and negative-time history is preserved. Each case sets $\beta_e(0) = \beta_e^* = z$, Fuzzy's percentage setpoint to $100z/\beta_{\max}$, and the marking to z plus the initial in-flight count. The two controllers share these values and the uncontrolled frequency vector. Capacity and matched gains remain fixed. Guards are enabled with zero reserve; observer, supervisory switching and anti-windup are disabled. We record complete-event buffer extrema, positive physical frequencies, limiter use, guard-off episodes and data progress. A guard-supported capacity invariant is distinguished from normal unstalled control.

Within each pair, capacity, compatible initialization, gain coordinates and execution policies are shared; the control law is the variable. Near-boundary targets test reserve requirements and are not proposed replacements for half-capacity centering. Table 8 maps each theoretical prediction to its verification and scope.

5.2 Reference safety and mismatch–occupancy relation

At $s = 1$, direct runs use $z \in \{16, 20, 24, 2024, 2028, 2032\}$. The nonzero-target executions reproduce the proved translation identity with maximum occupancy residual 6.83×10^{-13} frame and phase residual 2.78×10^{-12}. No comparison tolerance is loosened. At $z = 20$, Fuzzy's observed range is approximately $[2.685943, 37.314057]$, consistent with the certified all-time interval $(2.685, 37.315)$. PI's unprotected reference reaches -3.403701 on a receiving channel. Both controllers cross at $z = 16$ and both remain strictly safe at $z = 24$; the upper-target mirror checks agree under the zero-delay reference assumptions. Figure 1 shows the raw startup response. Negative occupancy after the first crossing is a diagnostic virtual continuation, never a physical buffer. The all-time conclusion comes from the prefix/tail certificate, not the finite simulation horizon.

The mismatch grid is $s \in \{0.25, 0.5, 0.75, 0.85, 0.86, 1, 1.125, 1.25, 1.5, 2\}$. Figure 2 separates the joint sufficient inner domain from finite-horizon observations at the frozen grid points. At $z = 20$, PI excursions of 19.893146 and 20.127183 frames at $s = 0.85$ and 0.86 bracket its proved directional capacity threshold. The Fuzzy excursions are computed directly, not scaled linearly. Successful sampled points do not certify the regions between them; failure of a sufficient test is recorded as uncertified.

For the uniform annex, an independent recomputation checks every one of the 20 parameter cells and reproduces the all-time bound below 17.342 over 399.6–400.4 ppm. Four direct endpoint runs are consistent: Fuzzy minima are 2.703366 and 2.668520, whereas PI minima are -3.380298 and -3.427105. Their role is to check the implementation and endpoints; full cell coverage and the contracting tail supply the uniform theorem.

Two additional zero-mean frequency directions complete the distribution check. Together with the prescribed direction, all have 400-ppm spread but $Q = (216, 234, 288)/30000^2$. Each satisfies the exact $V_1 = H^2 c_P Q$ relation. Their Fuzzy peaks differ (approximately 17.311137, 17.314057 and 17.322842 frames), while PI peaks coincide for these triangle cases. The energy relation is a sufficient-domain tool, not an exact-peak ordering rule.

5.3 Delayed Callisto: normal operation versus protection

The actual startup cases fix $s = 4$, or 1600-ppm initial spread, and $z \in \{64, 83, 104, 1944, 1965, 1984\}$. These assess whether the margin mechanism persists with local timing and integer frames; they do not apply the reference's numerical boundary to a different execution model. The low-target cases produce the predicted qualitative separation (Table 4). At $z = 83$, Fuzzy stays in $[11, 155]$ without intervention, while PI reaches zero and invokes five input-empty guard episodes. The first occurs at about 71844.672 physical ticks; summed blocked time is 61.0114 node-ticks. Figure 3 distinguishes physical clock motion from guarded data progress. At $z = 64$ both methods need protection; at $z = 104$ both remain normal. The advantage is therefore positive reserve without pausing the data execution at the same configuration. PI's guard prevents an invalid read; the observation is not a physical negative-buffer failure.

The upper-target results expose an additional configuration constraint. All six upper-target runs invoke outgoing-credit guards

Table 4: Actual low-target startup at matched gains. Ranges include every integer event; guard counts are off episodes, not off/on transitions.

	Buffer occupancy / frames		Guard episodes	
z	PI	Fuzzy	PI	Fuzzy
64	0–128	0–128	26	6
83	0–166	11–155	5	0
104	10–198	32–176	0	0

even though their maximum receiving occupancy is 2008, below $\beta_{\max} = 2048$. Equation (4) explains the additional constraint: initial forward-transit and unacknowledged-consumption counts are both 20, so used credit starts at $z + 40$, rather than at receiving occupancy z. Thus a simple physical-occupancy mirror of the zero-delay result does not predict guard-free delayed execution. This configuration-specific 40-frame observation is not a general latency margin.

All 12 runs retain inclusive physical capacity compliance and positive clock frequency (minimum 0.9993), without correction saturation. Nine cases rely on guards. Complete crossing replay has zero occupancy mismatches, and independent command reconstruction has maximum residual 1.09×10^{-19}. Pre-first-guard command and measurement prefixes match the old high-reserve trajectory exactly; all post-guard evolution is newly run. Fewer guard episodes alone does not prove higher throughput: at $z = 1965$ Fuzzy has fewer episodes but lower final data progress. Appendix D retains every case, credit diagnostics and the complete validation protocol.

5.4 Supplementary disturbance-performance evidence

Earlier independent tuning and random/burst campaigns answer a different question. Their complete protocols and results, including operating limits, are retained in Appendix E. They are not rerun or used as evidence for the safety-domain theorem. The present evidence chain is the mathematical configuration test, strict reference comparison and qualitative implementation checks.

6 Discussion

The analysis links a bittide storage requirement to an initial frequency mismatch through a specified controller and execution model. Mismatch energy $\delta\omega^\top L \delta\omega$ captures how frequency differences are distributed over the graph, while target placement determines the available lower and upper reserve. Sampling, gains, normalization and graph resistance enter the same joint test. These are coupled sufficient conditions; a failed test does not make a configuration unsafe, and maximum frequency spread alone does not specify the required reserve.

The matched-gain results isolate the implemented nonlinear law from changes in capacity, gain calibration or protection policy. They do not establish superiority over an independently optimized PI. The Fuzzy advantage has two distinct mathematical components. The exact nonlinear map explains state-dependent increment shaping; it does not itself order closed-loop performance. The outward prefix and contracting tail establish the strict excursion ordering.

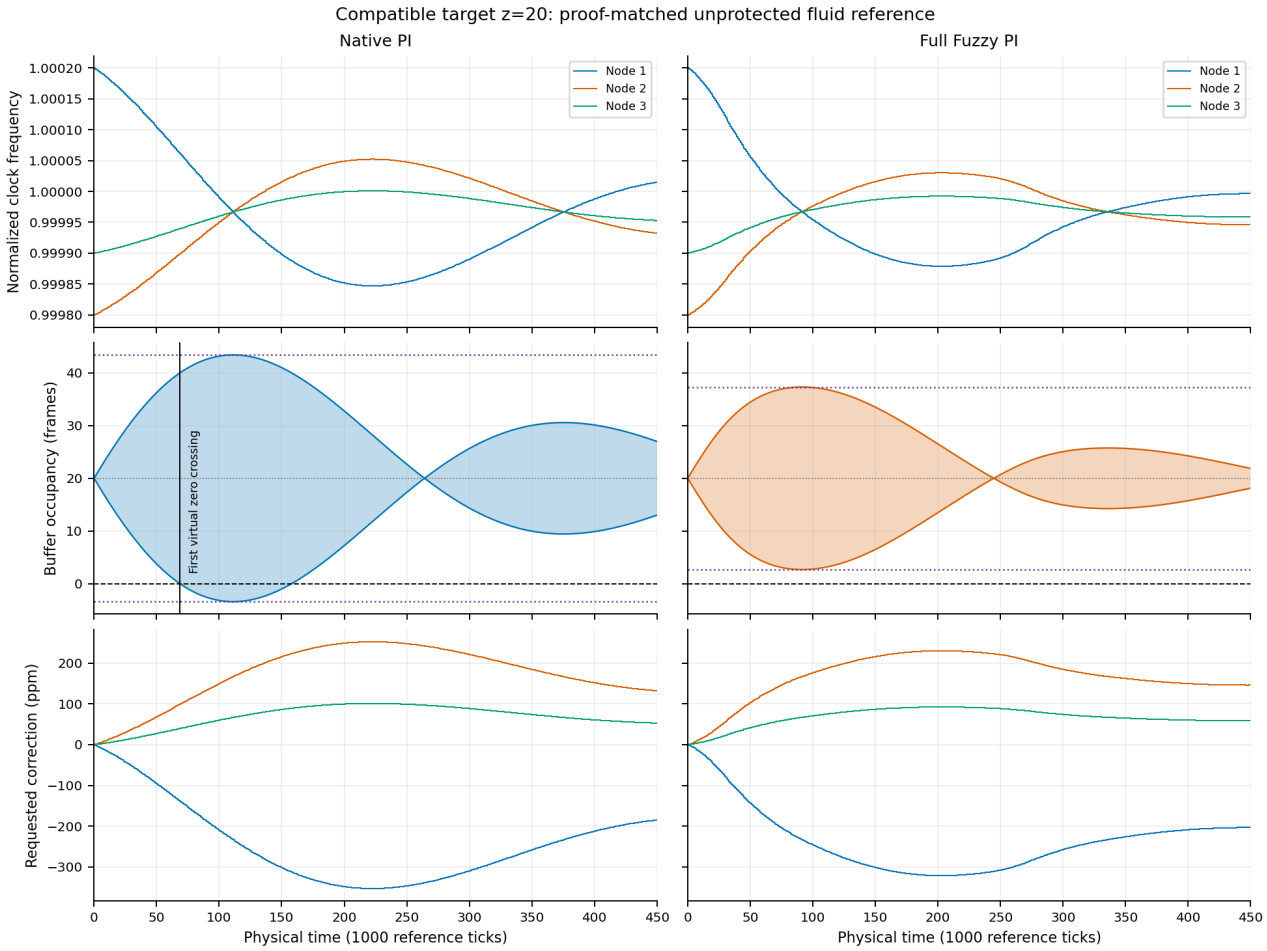


Figure 1: Proof-matched continuous fluid reference at $z = 20$ and 400-ppm initial spread. Occupancy bands cover all six channels. Fuzzy remains above zero, while the unprotected PI trajectory crosses the lower boundary. Dotted bounds come from the separate all-time certificate. Applied physical frequencies stay positive; signed corrections may be negative.

At fixed capacity the translation lemma then converts this ordering into a larger admissible target interval for a prescribed frequency initialization. Full parameter-cell coverage extends the result to a narrow, quantified interval along the same mismatch direction. This is a proved advantage for the stated family of configurations, with positive physical frequencies and no limiter intervention. It is not a claim that the coarser Lyapunov inner domain is larger for Fuzzy or that every 400-ppm initialization benefits.

The qualitative experiments directly check these deductions. Their cases are intentionally chosen around theoretical boundaries and are not independent statistical discovery. The reference uses the same continuous held flow as the proof. Callisto separately retains local scheduling, delay and integer events, so its normal-versus-guard separation supports the mechanism without transferring the reference boundary. The upper-target runs identify why receiving occupancy alone is insufficient for delayed readiness: used credit also includes forward-transit frames and consumption awaiting reverse feedback, as expressed by (4). A physical buffer below capacity can therefore coexist with a necessary guard. Fewer guard episodes can accompany lower final data progress.

The occupancy-translation family changes initial occupancy, target and compatible marking together. It does not certify arbitrary initial buffer error at a fixed half-capacity target, or preserve the original application's marking across configurations. Within each comparison the two controllers have the same marking and logical configuration. Keeping capacity fixed also preserves percentage normalization; changing capacity or topology requires consistent remapping and a fresh certificate.

The nominal reference proves target convergence and syntony with positive physical progress. Persistent delivery disturbances instead require finite-window performance measures. Earlier independently tuned experiments, retained in Appendix E, show benefits in two random regimes and costs in other regimes. They answer a separate tuning/performance question and do not enlarge the proved safe domain. Application-level logical synchrony also requires the prescribed frame relationships; buffer compliance and

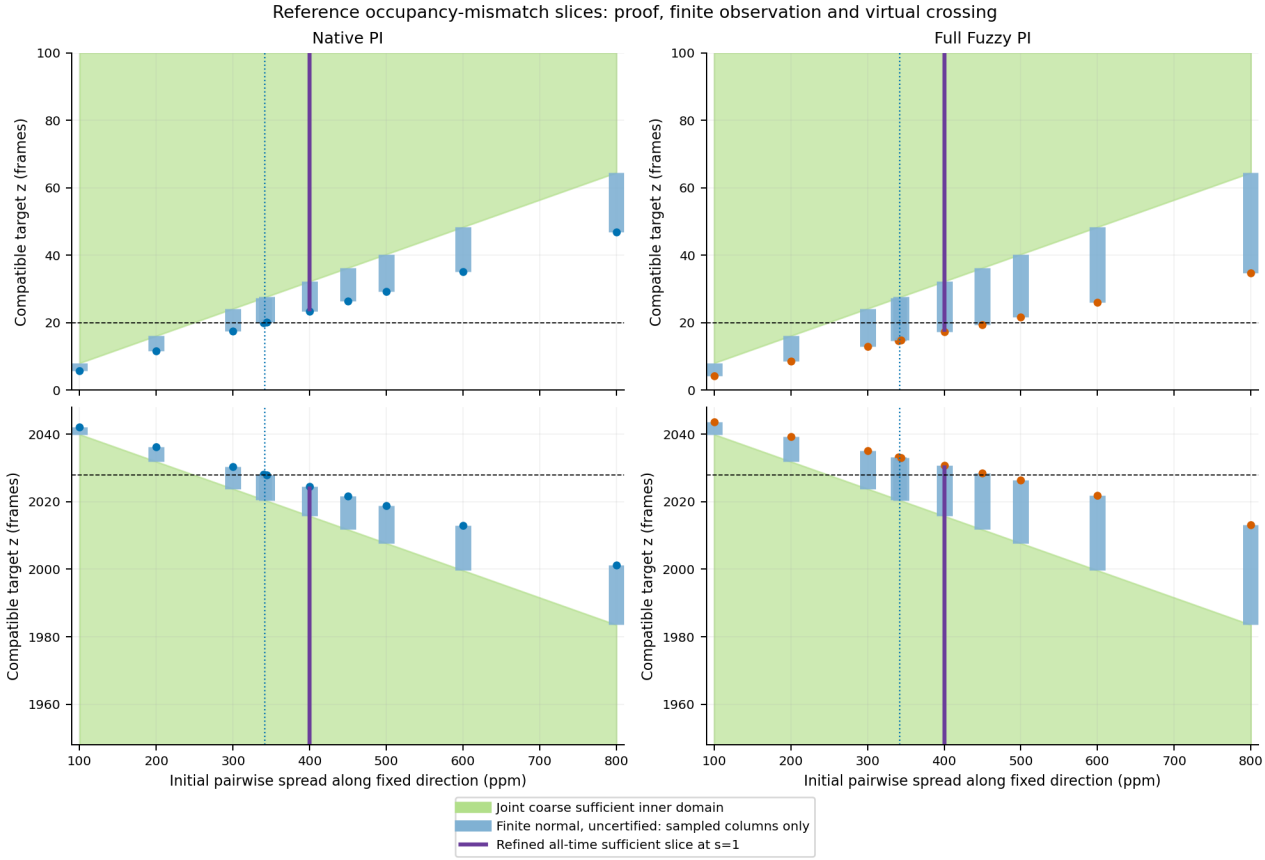


Figure 2: Compatible-target–mismatch slices at fixed capacity and gains. Green denotes the joint coarse sufficient domain. Blue columns extend to finite-horizon normal observations at sampled spreads only; markers give observed excursion boundaries. Purple denotes the sharper all-time slice at $s = 1$. The dotted line is PI's directional threshold at $z = 20$. Central targets are omitted for readability. No untested interpolation or delayed-Callisto safety boundary is asserted.

frequency agreement do not replace a payload refinement proof. Extending the invariant analysis to asynchronous sampling, nonzero link/control delay, integer counts and delayed credit is required before claiming all-time safety for the full Callisto implementation.

7 Conclusion

We proposed an incremental Fuzzy PI controller for bittide and derived configuration-dependent conditions connecting initial frequency mismatch, buffer reserve, graph structure and controller scales. Implementation-consistent PI calibration and the exact nonlinear rule map make the comparison explicit. For the stated synchronous zero-delay fluid reference, invariant-domain and Lyapunov analysis establish convergence, positive physical frequencies and correction bounds. A certified prefix and infinite tail prove a smaller Fuzzy buffer excursion and, at fixed capacity, a strictly larger interval of compatible safe occupancy targets for the prescribed initialization. Complete interval coverage additionally certifies a fixed-direction 399.6–400.4-ppm configuration region where Fuzzy is safe at target 20 while unprotected PI crosses zero.

Theory-directed simulations agree with these reference predictions. Twelve continuous-phase Callisto runs separately assess delayed integer execution. At target 83 and 1600-ppm initial spread, Fuzzy remains in $[11, 155]$ without protection, while PI reaches zero and invokes five input-empty guard episodes. Upper-target cases expose the additional delayed-credit condition and delimit the reference mirror argument. The result is a conditional mathematical safety advantage supported by qualitative implementation evidence, with all-time claims restricted to the declared reference. Incorporating asynchronous timing, delay, quantization and credit into a full execution certificate remains necessary for broader guarantees.

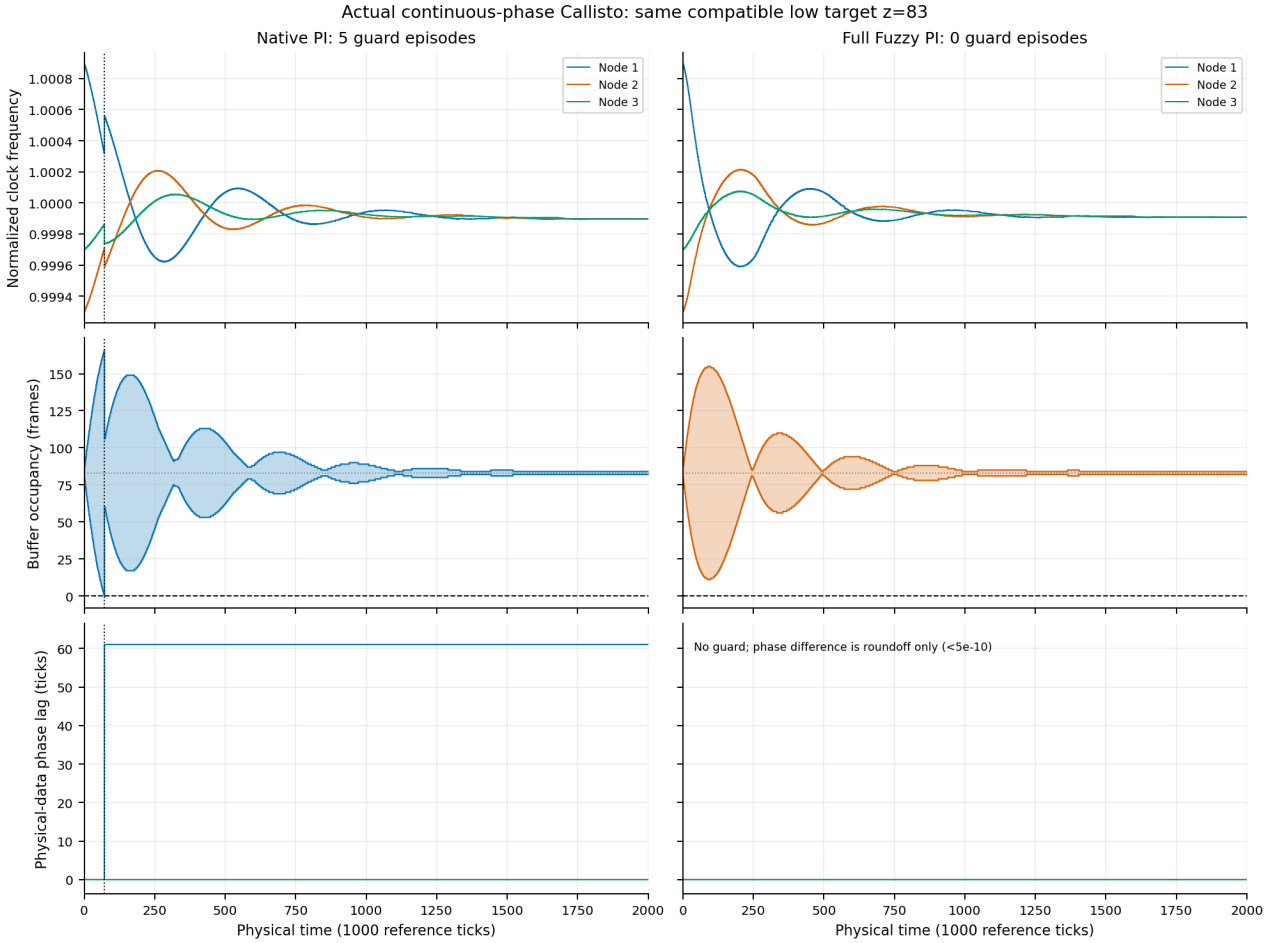


Figure 3: Continuous-phase Callisto at $z = 83$, 1600-ppm spread and matched gains. Full-event occupancy envelopes show PI reaching zero and Fuzzy retaining positive reserve. The dotted line marks PI's first guard. Physical-minus-data phase reveals guard-induced progress loss; both columns use common axis ranges. This finite delayed execution supports the mechanism and is separate from the zero-delay all-time proof.

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A Extended Background

Correspondence: Section 2. This appendix preserves the extended model, intermediate steps and supporting interpretation behind the corresponding main-text section.

A.1 Logical Synchrony and the bittide Mechanism

Logical synchrony describes relationships between distributed events without requiring a global physical clock. In bittide, neighboring nodes exchange ordered frames, and elastic receiving buffers accommodate differences in their physical progress. The association between sending and receiving events provides the basis for a logical execution whose dependencies can be reasoned about independently of exact wall-clock timing. Equal physical phases are therefore not required for logical synchrony. Frequency agreement alone, however, does not establish the required event relationships: frame ordering, initialization and continued availability of the corresponding frames also matter. These distinctions follow the logical-synchrony and multiclock-network framework developed by Lall et al. [3].

The control problem supports this realization by regulating physical progress while retaining the prescribed frame relationships. Each node observes its own receiving buffers and adjusts its local clock frequency. Buffers consequently provide both the local feedback signal and finite storage for a temporary imbalance between delivery and consumption.

A.2 Continuous Clocks and Receiving-Buffer Dynamics

We consider a network of n nodes. Node i has continuous clock phase $\theta_i(t)$, measured in local cycles, and physical frequency

$$\dot{\theta}_i(t) = \omega_i(t) = \omega_i^u + c_i(t), \tag{25}$$

where ω_i^u is its constant uncontrolled frequency and $c_i(t)$ is the applied frequency correction. The correction can be signed; admissible physical operation requires $\omega_i(t) > 0$. In the current Callisto model, corrections are held between local actuation events. Phase remains continuous when a correction changes, although its slope can change. Local sampling and integer frame events are embedded in this continuous-time plant.

For a receiving edge $e = (j \to i)$, let $\beta_e(t)$ be buffer occupancy in frames and $\beta_{\max,e}$ its capacity. During normal operation without progress blocking,

$$\begin{aligned} \beta_e(t) &= \beta_e(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \\ N_i(t) &= \lfloor \theta_i(t) \rfloor - \lfloor \theta_i(0) \rfloor. \end{aligned} \tag{26}$$

Here $A_{ji}^{\mathrm{arr}}(t)$ counts arrivals during $(0, t]$, including frames already in flight at initialization, and $N_i(t)$ counts local consumption. This conservation relation explains why both clock frequency mismatch and delivery changes affect occupancy. Section B specifies the local schedules, initialization and ordered arrival model used in our realization.

Fluid models replace integer frame counts by continuous phase increments. With constant physical link latency τ_{ji}, the corresponding occupancy satisfies, between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j(t - \tau_{ji}) - \omega_i(t), \tag{27}$$

where $\widetilde{\beta}_{ji}$ denotes fluid occupancy. Thus occupancy integrates the imbalance between delayed source progress and local consumption. The fluid approximation facilitates analysis; its assumptions must be distinguished from those of integer-frame execution. This separation is also central to existing PI performance analysis [1] and buffer-occupancy modeling [4].

A.3 Safety, Syntony and Buffer Centering

Safety is a property of the complete trajectory. For the normal operating regime studied here, we seek

$$0 < \beta_e(t) < \beta_{\max,e}, \qquad \omega_i(t) > 0, \qquad |c_i(t)| \le c_{\max}, \tag{28}$$

for every edge, node and time, where $c_{\max} > 0$ is the correction limit. The available lower and upper buffer reserves are $\beta_e(t)$ and $\beta_{\max,e} - \beta_e(t)$, respectively. A convergent trajectory can still exhaust either reserve before reaching its limit. Avoiding such exhaustion through control is distinct from invoking a protection mechanism that blocks logical progress.

Define the physical frequency spread as

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{29}$$

For nominal fluid operation with fixed conditions, we seek $\Delta_\omega(t) \to 0$, together with $\beta_e(t) \to \beta_e^*$ for compatible occupancy targets β_e^*. Frequency agreement is the nominal syntonization objective; it does not imply zero phase differences or tracking an externally prescribed frequency. Without an external reference, the common limiting frequency ω_∞ is determined by the closed loop. Likewise, arbitrary per-edge targets cannot be assumed compatible with a fixed marking, meaning the frame-count offsets determined by initialization, and with the selected initial state.

If a transient frequency-spread tolerance $\Delta_{\mathrm{safe}} > 0$ is specified, then $\Delta_\omega(t) < \Delta_{\mathrm{safe}}$ is an additional engineering constraint. It is not the definition of logical synchrony. Under persistent delivery perturbations, we assess bounded deviations, residual frequency differences and buffer reserves rather than assume exact disturbance-free centering.

A.4 Delivery Perturbations and Local Feedback

A latency increase affects the delivery process before a receiver necessarily observes it through its next control measurement. If arrivals pause while consumption continues, the conservation relation shows that the receiving buffer decreases; subsequent release of accumulated frames can instead increase occupancy. The disturbance therefore acts through frame availability and accumulated count imbalance. A controller cannot recreate an undelivered frame, and its response is constrained by measurement and actuation timing. Our experiments use declared order-preserving delivery traces so that controller comparisons share the same external stress definition. The exact arrival construction is given in Section B.

A.5 Existing Control Foundations and the Present Study

Existing work analyzes occupancy-feedback PI performance through controller gains and graph resistance distances, derives limiting fluid occupancies, and develops reframing methods that separate frequency regulation from occupancy positioning. These are complementary foundations: PI performance [1], steady buffer occupancy [4], and buffer centering [2]. Their conclusions apply under their respective modeling assumptions; in particular, continuous-time gain conditions cannot be transferred unchanged to sampled execution with delay.

This paper builds on those foundations by proposing a local incremental Fuzzy PI law and analyzing how its nonlinear increments reshape the closed-loop response at explicitly calibrated PI scales. Section B defines the complete controller and establishes that calibration. Section C derives sufficient safety and convergence conditions for a declared synchronous, zero-latency fluid reference, followed by an all-time buffer-excursion comparison for one prescribed initialization. Evaluation in continuous-phase Callisto separately examines delivery perturbations under the actual local update and frame-count conventions.

B Controller Realization and Complete Derivations

Correspondence: Section 3. This appendix preserves the extended model, intermediate steps and supporting interpretation behind the corresponding main-text section.

We propose a decentralized, occupancy-based incremental Fuzzy PI controller for bittide. Each node combines receiving-buffer error with its update-to-update change to compute an frequency correction increment. This section specifies the complete controller and its continuous-clock interface, derives an implementation-consistent PI calibration, and establishes the nonlinear properties used in Section C. The physical plant, integer frame counts and sampled controller state are kept distinct throughout the derivation.

B.1 Continuous-clock realization of the proposed controller

The plant is the current continuous-phase Callisto model within the bittide framework [3]. For constant uncontrolled clock frequency ω_i^u and applied correction $c_i(t)$,

$$\begin{aligned}\dot{\theta}_i(t) &= \omega_i^u + c_i(t),\\ c_i(t) &= c_{i,k}, \qquad t \in [t_{i,k}^a, t_{i,k+1}^a),\\ \theta_i(t) &= \theta_i(t_{i,k}^a) + (\omega_i^u + c_{i,k})(t - t_{i,k}^a).\end{aligned} \tag{30}$$

The superscript a denotes actuation, and k indexes that node's updates. Normal executions require $\omega_i^u + c_i(t) > 0$, so the phase has a monotone inverse. Physical phase is continuous at an update; its slope can change. Requested correction $q_{i,k}$ is computed from the measured buffers and applied as $c_{i,k} = \mathrm{sat}_{c_{\max}}(q_{i,k})$, with $\mathrm{sat}_b(z) = \min\{b, \max\{-b, z\}\}$ and $c_{\max} > 0$. The proposed controller has jump updates inside a continuous physical plant; it is not implemented as an ordinary differential equation for its accumulator.

Let $p > 0$ be the polling period and $d_{\mathrm{ctrl}} \geq 0$ the execution delay, both measured in local-clock phase units. Measurement time $t_{i,k}^m$ and actuation time $t_{i,k}^a$ satisfy

$$\begin{aligned}\theta_i(t_{i,k}^m) &= \theta_i(0) + k\,p,\\ \theta_i(t_{i,k}^a) &= \theta_i(0) + k\,p + d_{\mathrm{ctrl}},\\ \int_{t_{i,k}^m}^{t_{i,k}^a} (\omega_i^u + c_i(s))\,ds &= d_{\mathrm{ctrl}}.\end{aligned} \tag{31}$$

The product $k\,p$ in these scheduling equations means the integer index k multiplied by the poll period p; it is not the proportional gain k_p. With zero initial correction, $t_{i,0}^m = 0$ and $t_{i,0}^a = d_{\mathrm{ctrl}}/\omega_i^u$. Subsequent actuation intervals equal $p/(\omega_i^u + c_{i,k})$. The delay is not automatically a constant physical time, and its age must not be computed using the new command that is applied only at the end of that age interval.

In normal operation, each integer source-phase crossing sends a frame. With the progress guards open, frame conservation gives

$$\beta_{j\to i}(t) = \beta_{j\to i}(0) + A_{ji}^{\mathrm{arr}}(t) - N_i(t), \tag{32}$$

where $A_{ji}^{\mathrm{arr}}(t)$ counts arrivals during $(0, t]$, including initial in-flight frames, and $N_i(t) = \lfloor\theta_i(t)\rfloor - \lfloor\theta_i(0)\rfloor$ counts local consumption. For a constant physical delay $\tau_{ji} \geq 0$,

$$A_{ji}^{\mathrm{arr}}(t) = \lfloor\theta_j(t - \tau_{ji})\rfloor - \lfloor\theta_j(-\tau_{ji})\rfloor. \tag{33}$$

Thus arrival counts incorporate the declared negative-time history.

For constant physical link latency τ_{ji}, receiving occupancy is

$$\beta_{j\to i}(t) = \lambda_{ji}^{\mathrm{UGN}} + \lfloor\theta_j(t - \tau_{ji})\rfloor - \lfloor\theta_i(t)\rfloor. \tag{34}$$

Here $\lambda_{ji}^{\mathrm{UGN}}$ is the fixed integer offset in the absolute phase-count convention. It is determined by occupancy and prehistory:

$$\lambda_{ji}^{\mathrm{UGN}} = \beta_{j\to i}(0) - \lfloor\theta_j(-\tau_{ji})\rfloor + \lfloor\theta_i(0)\rfloor. \tag{35}$$

The relative-count logical marking is $\lambda_{ji}^{\mathrm{LSN}} = \lambda_{ji}^{\mathrm{UGN}} + \lfloor\theta_j(0)\rfloor - \lfloor\theta_i(0)\rfloor$; the two coincide when the initial phase counts are equal.

For changing delivery times, let $s_{j,m} = \theta_j^{-1}(m)$ be the send time of source tick m, and $a_{ji}(s)$ its causal ordered arrival map. For the desired send-time delay profile $\tau_{ji}(s)$, the declared running-maximum envelope is

$$a_{ji}(s) = \sup_{r\leq s}\{r + \tau_{ji}(r)\},$$

including the constant-delay negative-time history. Define the delivered tick frontier

$$\begin{aligned}M_{ji}^{\mathrm{arr}}(t) &= \max\{m \in \mathbb{Z} : a_{ji}(s_{j,m}) \leq t\},\\ \beta_{j\to i}(t) &= \lambda_{ji}^{\mathrm{UGN}} + M_{ji}^{\mathrm{arr}}(t) - \lfloor\theta_i(t)\rfloor.\end{aligned} \tag{36}$$

The negative-time history specifies initial in-flight frames. The declared running-maximum arrival envelope preserves order and can release a batch after a desired-delay decrease; it imposes no finite serialization rate. At an upward-delay discontinuity, a supremum of continuous source times need not itself have arrived. Counting delivered tick indices avoids silently including such a boundary frame.

If a protection guard blocks logical progress, the runtime can retain a separate continuous data phase ψ_i with $\dot{\psi}_i = g_i(t)\dot{\theta}_i$, where $g_i \in \{0, 1\}$ is the progress gate. Buffer counts then use ψ_i, whereas

scheduling still uses θ_i. In the ungated normal regime, with matching initial phases, data phase and physical phase agree. Guard operation does not stop the oscillator.

Only for the reference proof do we replace integer occupancy by a fluid counterpart. At constant delay, define $\widetilde{\beta}_{ji}(t) = \widetilde{\lambda}_{ji} + \theta_j(t - \tau_{ji}) - \theta_i(t)$, where $\widetilde{\lambda}_{ji}$ is an additive offset chosen for the specified fluid initialization. Between command discontinuities,

$$\dot{\widetilde{\beta}}_{ji}(t) = \omega_j^u + c_j(t - \tau_{ji}) - \omega_i^u - c_i(t). \tag{37}$$

This derivative belongs to the fluid counterpart, not the actual integer occupancy. Replacing floor counts by phases does not automatically preserve an arbitrary initialization; the compatible zero-latency marking used in Section C is specified there.

B.2 Notation and the measured error

Node i has incoming-neighbor set $\mathcal{N}_i^-$ and $d_i = |\mathcal{N}_i^-| > 0$ incoming channels. All incoming sums below range over $j \in \mathcal{N}_i^-$. This section assumes fixed active incoming sets: the runtime link-status mask is one on all configured inputs. The Fuzzy percentage wrapper averages its configured inputs, so changing link masks would require an explicit corresponding measurement policy. At the kth local measurement, $\beta_{ji,k}$ is the measured occupancy of channel $j \to i$, and β_{ji}^* is its fixed target. Use

$$E_{i,k} = \sum_{j \in \mathcal{N}_i^-} (\beta_{ji,k} - \beta_{ji}^*). \tag{38}$$

The sign is actual minus target. If incoming buffers are too full, a positive frequency correction increases local consumption. This explains the positive PI gains with this error convention. The physical frequency is $\omega_i = \omega_i^u + c_i$, where ω_i^u is the constant uncontrolled frequency; c_i is a correction, not the entire frequency.

Symbol	Meaning
$E_{i,k}$	Sum error in frames
$e_{i,k}$	Target-minus-measured error in percentage points
$I_{i,k}$	PI baseline sum of sampled frame errors
$A_{i,k}$	Fuzzy accumulator: total requested correction
$q_{i,k}, c_{i,k}$	Requested and applied frequency corrections
$c_{\max}$	Magnitude limit on the applied correction
$\beta_{\max}$	Common buffer capacity in frames
$\beta_{\max,ji}$	Capacity of receiving channel $j \to i$
χ_{ji}	Used outgoing credit in frames
h	Fixed PI baseline integration scale
$X_{i,k}, Y_{i,k}$	Scaled Fuzzy error and sample-difference inputs
$F_0, \mathcal{R}_F$	Linear counterpart and residual $F - F_0$

The event index k is local to a node. The normalization factor γ_i converts a frame-sum error into percentage points. The maximum correction $c_{\max} > 0$ is the actuator limit; q and c remain distinct if clipping acts. Two nodes need not measure at the same physical times. The subscript ji abbreviates edge $j \to i$. Suppressing i below is a notational convenience, not a synchrony assumption; in that notation $\gamma = \gamma_i$.

B.2.1 Deriving the percentage-to-frame-error relation. For common incoming capacity $\beta_{\max}$, define

$$\bar{p}_{i,k} = \frac{1}{d_i} \sum_j \frac{100\beta_{ji,k}}{\beta_{\max}}, \qquad r_i = \frac{1}{d_i} \sum_j \frac{100\beta_{ji}^*}{\beta_{\max}}.$$

Then subtraction and factoring give

$$e_{i,k} = r_i - \bar{p}_{i,k} \tag{39}$$

$$= -\frac{100}{d_i \beta_{\max}} \sum_j (\beta_{ji,k} - \beta_{ji}^*) = -\gamma_i E_{i,k}, \qquad \gamma_i = \frac{100}{d_i \beta_{\max}}. \tag{40}$$

For midpoint targets this is the original absolute-percentage input with $r_i = 50$. For other targets the runtime can supply a centered percentage $r_i + d_i^{-1} \sum_j 100(\beta_{ji,k} - \beta_{ji}^*)/\beta_{\max}$; subtracting it from r_i yields the same $e_{i,k}$. A mismatched setpoint instead adds a constant bias and invalidates the scalar relation used for calibration.

With unequal capacities and a consistently matched target reference, the percentage average gives

$$e_{i,k} = -\frac{100}{d_i} \sum_j \frac{\beta_{ji,k} - \beta_{ji}^*}{\beta_{\max,ji}}.$$

Here $\beta_{\max,ji}$ is the capacity of receiving channel $j \to i$. This is a weighted error and generally cannot be written as one normalization constant times the unweighted PI error sum. Matching such a controller requires a consistent measurement weighting or a different baseline.

B.3 PI baseline: update order, units and the first call

The occupancy-feedback PI baseline is grounded in the bittide analysis of [1]. The implemented PI update is an integrate-then-output recursion:

$$I_k = I_{k-1} + E_k, \qquad q_k^{\mathrm{PI}} = k_p E_k + \kappa_I I_k, \qquad \kappa_I = k_i h. \tag{41}$$

The sum I_k contains no physical time multiplier; h is incorporated in κ_I. In the present runtime,

$$h = \frac{p}{\omega_{\mathrm{base}}}. \tag{42}$$

Here p is the local polling period and ω_{base} is the fixed implementation reference frequency. The scale h stays fixed during the configured run. Local polling intervals measured in physical time can vary as the local clock frequency changes; replacing h by those intervals would implement a different PI law.

Subtract the previous requested output:

$$q_k^{\mathrm{PI}} - q_{k-1}^{\mathrm{PI}} = k_p(E_k - E_{k-1}) + \kappa_I(I_k - I_{k-1}) \tag{43}$$

$$= k_p(E_k - E_{k-1}) + \kappa_I E_k. \tag{44}$$

This is the correct increment to reproduce. It contains the present E_k in the integral term because integration precedes output.

At zero initial sum, the first call returns

$$q_0^{\mathrm{PI}} = (k_p + \kappa_I)E_0. \tag{45}$$

No physical prehistory measurement is needed to express this algebra: one may use the fictitious previous error $E_{-1} = 0$ and previous requested output $q_{-1} = 0$. That bookkeeping choice is separate from deriving a plant identity such as $x_k - x_{k-1} = Hw_{k-1}$, which requires a genuine elapsed plant interval.

Table 5: Rule consequents of the proposed bittide Fuzzy PI controller.

$X\backslash Y$	NL	NS	ZE	PS	PL
NL	NL	NL	NL	NS	ZE
NS	NL	NL	NS	ZE	PS
ZE	NL	NS	ZE	PS	PL
PS	NS	ZE	PS	PL	PL
PL	ZE	PS	PL	PL	PL

B.4 The proposed Fuzzy PI controller

The proposed local incremental control law combines the present error and its sample difference as follows:

$$X_k = K_e e_k, \qquad Y_k = K_{\Delta e}(e_k - e_{k-1}), \tag{46}$$

$$A_k = A_{k-1} + K_u F(X_k, Y_k), \qquad q_k^{\mathrm{F}} = A_k. \tag{47}$$

The gain signs used here are $K_e, K_{\Delta e} > 0$ and $K_u < 0$. Thus high occupancy produces negative e, and a negative surface output produces a positive correction increment. The quantity $e_k - e_{k-1}$ is not divided by elapsed time; calling it a time derivative would obscure its sampling dependence.

B.4.1 Memberships and rule weights. The labels NL, NS, ZE, PS and PL mean negative large, negative small, zero, positive small and positive large, respectively. Each membership $\varphi_a(z) \in [0, 1]$ is an activation degree; a singleton is a numerical rule consequent. Let $[z]_+ = \max\{z, 0\}$ and $\mathrm{clip}_{[0,1]}\, z$ denote projection onto $[0, 1]$. The exact memberships can be written

$$\begin{aligned} \varphi_{\mathrm{NL}}(z) &= \mathrm{clip}_{[0,1]}\,\frac{-z-5}{5}, & \varphi_{\mathrm{NS}}(z) &= [1-|z+5|/5]_+, \\ \varphi_{\mathrm{ZE}}(z) &= [1-|z|/5]_+, & \varphi_{\mathrm{PS}}(z) &= [1-|z-5|/5]_+, \\ \varphi_{\mathrm{PL}}(z) &= \mathrm{clip}_{[0,1]}\,\frac{z-5}{5}. \end{aligned} \tag{48}$$

Table 5 gives all 25 consequent rules. Order the labels as NL, NS, ZE, PS and PL, with label indices $a, b \in \{1, \ldots, 5\}$. With singleton values $(s_1, \ldots, s_5) = (-5, -2.5, 0, 2.5, 5)$ and $\ell(a, b) = \min\{5, \max\{1, a+b-3\}\}$, compute

$$w_{ab} = \min\{\varphi_a(X), \varphi_b(Y)\}, \qquad F(X, Y) = \frac{\sum_{a,b} w_{ab} s_{\ell(a,b)}}{\sum_{a,b} w_{ab}}. \tag{49}$$

The membership partition gives at least one positive input membership on each axis, hence a positive denominator at finite inputs. Individual rules are summed even when they share a consequent. Replacing this accumulation by maximum aggregation changes the map and its derivation.

The weighted-average property gives $|F| \le 5$ and therefore $|A_k - A_{k-1}| \le 5|K_u|$. This is a per-update bound, not a bound on A_k over an arbitrarily long execution. The raw Fuzzy module itself has no final correction clipping. Initialize $A_{i,-1} = 0$ and $e_{i,-1} = 0$. At each measurement, form $e_{i,k}$ and $(X_{i,k}, Y_{i,k})$, evaluate the rule weights and normalized surface, increment $A_{i,k}$, and store $e_{i,k}$ as the previous error. The requested correction is then limited and applied at $t_{i,k}^a$, and held until the next actuation. In the normal-run policy considered here, external clipping retains the raw accumulator and anti-windup is disabled. The controller does not overwrite $A_{i,k}$ with the applied command. The membership functions, rule consequents and gains stay fixed during a run; the correction changes with the measured input pair. The law requires neither an observer nor an FFP mode.

B.5 Exact PI Calibration, Including Initialization

Define the globally linear counterpart

$$F_0(X, Y) = \frac{X + Y}{2}. \tag{50}$$

Replacing F by this map retains the same controller accumulator. Using $e = -\gamma E$ gives

$$A_k^0 - A_{k-1}^0 = \frac{K_u}{2}\{K_e e_k + K_{\Delta e}(e_k - e_{k-1})\} \tag{51}$$

$$= -\frac{K_u \gamma K_e}{2} E_k - \frac{K_u \gamma K_{\Delta e}}{2}(E_k - E_{k-1}). \tag{52}$$

Matching the two independent coefficients in (44) yields

$$k_p = -\frac{K_u \gamma K_{\Delta e}}{2}, \qquad \kappa_I = -\frac{K_u \gamma K_e}{2}. \tag{53}$$

Solving for two Fuzzy gains with chosen $K_e > 0$ gives

$$K_u = -\frac{2k_i h}{\gamma K_e}, \qquad K_{\Delta e} = \frac{K_e k_p}{k_i h}. \tag{54}$$

The formulas assume $k_i h > 0$. Pure proportional control is a separate limiting case and should not be inserted into this inverse mapping with a zero denominator.

Proposition B.1 (Event-level linear-counterpart equivalence). *For any measurement sequence, suppose* (53) *holds and initialize*

$$A_{-1}^0 = k_p E_{-1} + \kappa_I I_{-1}, \qquad e_{-1} = -\gamma E_{-1}. \tag{55}$$

Then $A_k^0 = k_p E_k + \kappa_I I_k$ *for every* $k \ge 0$.

Proof. Assume the relation holds before update k. Substitution gives

$$\begin{aligned} A_k^0 &= k_p E_{k-1} + \kappa_I I_{k-1} + k_p(E_k - E_{k-1}) + \kappa_I E_k \\ &= k_p E_k + \kappa_I (I_{k-1} + E_k) = k_p E_k + \kappa_I I_k. \end{aligned}$$

The stated initialization supplies the induction base, including $k = 0$. □

B.5.1 Why zero constructors work for a nonzero first error. Take $I_{-1} = A_{-1}^0 = e_{-1} = 0$, corresponding to $E_{-1} = 0$. The first Fuzzy-reference input is

$$(X_0, Y_0) = (-\gamma K_e E_0, -\gamma K_{\Delta e} E_0),$$

so $A_0^0 = (k_p + \kappa_I) E_0$, exactly as in (45). If only Fuzzy is warmed up to $e_{-1} = e_0$ while $A_{-1}^0 = 0$, its difference input vanishes and the output is merely $\kappa_I E_0$. The missing $k_p E_0$ term can create a large apparent controller difference unrelated to fuzzy inference.

B.5.2 A numerical calibration from the present configuration. For $d = 2$, $\beta_{\max} = 2048$, $p = 1000$, $\omega_{\text{base}} = 29999/30000$, $k_p = 2 \times 10^{-6}$, $k_i = 5 \times 10^{-11}$ and $K_e = 0.1$,

$$\gamma = \frac{25}{1024}, \qquad h = \frac{30000000}{29999},$$
$$\kappa_I = \frac{3}{59998000}, \qquad K_u = -\frac{768}{18749375},$$
$$K_{\Delta e} = \frac{29999}{7500} \simeq 3.9998666667. \tag{56}$$

For a one-frame first error, the matched PI and F_0 corrections are both $2.0500016667 \times 10^{-6}$. Warming up only the Fuzzy previous error instead gives approximately $5.0001666722 \times 10^{-8}$. The example illustrates an initialization mismatch; it is not a new plant simulation.

B.5.3 Execution-layer conditions. Identical raw outputs give identical applied outputs under a common limiter when raw states are retained and anti-windup is disabled. Tracking or reset requires compatible state changes. For example, after processing E_k, setting

$$A_k^0 = c_{\text{held}}, \qquad I_k = \frac{c_{\text{held}} - k_p E_k}{\kappa_I}, \qquad e_k = -\gamma E_k$$

restores the state relation for the next ordinary update. A branch designed to produce a held output on an immediate extra update requires its own integrate-first adjustment; the above is not an instruction to add an extra call. Limiting the internal reference F_0 to $[-5, 5]$ also changes the controller, even if the external correction limiter is shared.

Under a deterministic causal plant, the same initial history, measured inputs, actuation order, external trace and compatible resets imply closed-loop equality by induction over events. No globally synchronous physical grid is required for this algebraic equivalence. By contrast, matching only labels such as "same gains" or "same random seed" does not establish these conditions.

B.6 Deriving the Exact Central Nonlinear Surface

For $0 < r \le 5$, define

$$\Omega_r = \{(X, Y) : |X| + |Y| \le r\},$$
$$m = \min(|X|, |Y|), \qquad s = |X| + |Y|. \tag{57}$$

The same-sign interaction is

$$J(X, Y) = \begin{cases} \operatorname{sgn}(X) m, & XY > 0, \\ 0, & XY \le 0. \end{cases} \tag{58}$$

For points in Ω_5, put $a = |X|/5$, $b = |Y|/5$. Because $a + b \le 1$, each input has only its central and sign-adjacent membership active. Their four joint min weights are

$$w_{00} = \min(1 - a, 1 - b) = 1 - \max(a, b),$$
$$w_{s0} = \min(a, 1 - b) = a,$$
$$w_{0s} = \min(1 - a, b) = b,$$
$$w_{ss} = \min(a, b). \tag{59}$$

The subscripts s denote the input's active sign label, not the scalar $s = |X| + |Y|$. Zero-input boundaries follow by continuity.

The denominator is therefore

$$D = 1 - \max(a, b) + a + b + \min(a, b) = 1 + 2\min(a, b) = 1 + 2m/5. \tag{60}$$

The central-central consequent is zero. The sign-central and central-sign rules contribute $(5/2)\operatorname{sgn}(X)a = X/2$ and $(5/2)\operatorname{sgn}(Y)b = Y/2$. For equal signs, the joint consequent is $5\operatorname{sgn}(X)$ and contributes $\operatorname{sgn}(X)m$; for opposing signs it is zero. The four quadrants therefore give

$$F(X, Y) = \begin{cases} \dfrac{F_0 + m}{1 + 2m/5}, & X > 0,\ Y > 0, \\ \dfrac{F_0 - m}{1 + 2m/5}, & X < 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X > 0,\ Y < 0, \\ \dfrac{F_0}{1 + 2m/5}, & X < 0,\ Y > 0. \end{cases} \tag{61}$$

On either axis $m = 0$, giving $F = F_0$. These cases establish the following identity, including boundaries by continuity.

Proposition B.2 (Exact central surface). *For every* $(X, Y) \in \Omega_5$,

$$F(X, Y) = \frac{(X + Y)/2 + J(X, Y)}{1 + 2m/5},$$
$$J(X, Y) = \begin{cases} \operatorname{sgn}(X) m, & XY > 0, \\ 0, & XY \le 0. \end{cases} \tag{62}$$

This is an exact rule-level identity on Ω_5, not a fitted approximation. Outside this diamond, the cross weights in (59) need not equal a and b, so the formula must not be extended without a new case analysis.

B.7 Sign-conditioned response, sector and nonlinear geometry

B.7.1 Same-sign enhancement. When $XY > 0$, F_0 has magnitude $s/2$ and the same sign as X. Equation (62) gives

$$\frac{F}{F_0} = \frac{1 + 2m/s}{1 + 2m/5}. \tag{63}$$

Since $s \le 5$, its numerator is at least the denominator. Since $m \le s/2$, the numerator is at most 2. Thus $1 \le F/F_0 \le 2$. Define the normalized nonlinear residual by $\mathcal{R}_F(X, Y) = F(X, Y) - F_0(X, Y)$. Subtracting F_0 gives

$$\mathcal{R}_F = \frac{\operatorname{sgn}(X) m (1 - s/5)}{1 + 2m/5}, \qquad XY > 0. \tag{64}$$

For instance, $F(0.1, 0.1) = 0.2/1.04 \simeq 0.19230769$, compared with $F_0 = 0.1$. Both are surface values; the actual increment multiplies by negative K_u.

B.7.2 Opposing-sign attenuation and zero-output cases. When $XY < 0$, the interaction vanishes, so

$$F = \frac{F_0}{1 + 2m/5}, \qquad \mathcal{R}_F = -\frac{2m/5}{1 + 2m/5} F_0. \tag{65}$$

For $(X, Y) = (0.1, -0.05)$, $F_0 = 0.025$ and $F = 0.025/1.02 \simeq 0.024509804$. On either axis $m = 0$ and $F = F_0$. On the anti-diagonal $Y = -X$, both outputs are exactly zero; a ratio should not be evaluated there.

Corollary B.3 (Regional pointwise sector). *For $(X, Y) \in \Omega_r$, $0 < r \le 5$, one can write*

$$F = \alpha(X, Y)F_0, \qquad \underline{\alpha}_r = \frac{1}{1 + r/5} \le \alpha(X, Y) \le 2. \tag{66}$$

Proof. Same-sign ratios lie in $[1, 2]$. For opposing signs with $F_0 \neq 0$, $m \le r/2$ gives $1/(1 + r/5) \le F/F_0 \le 1$. The axes agree with F_0. If $F_0 = 0$, both outputs vanish and any admissible α preserves the equality. □

The equivalent division-free inequality is

$$[F - \underline{\alpha}_r F_0][2F_0 - F] \ge 0. \tag{67}$$

In particular, Ω_1 gives $\alpha \in [5/6, 2]$, while Ω_5 gives $[1/2, 2]$. These bounds describe pointwise values of the specific nonlinear map. They are not a gain-scheduled stability proof; the closed loop must retain the input region and accommodate different node-dependent, time-varying α values.

B.7.3 Why an axis match is not a Jacobian linearization. The same-sign term J is positively homogeneous. For $t > 0$ and a fixed direction $v = (v_1, v_2)$, the central formula yields

$$F'(0; v) = \lim_{t \downarrow 0} \frac{F(tv_1, tv_2)}{t} = \frac{v_1 + v_2}{2} + J(v_1, v_2). \tag{68}$$

Consequently $F'(0; (1, 0)) = F'(0; (0, 1)) = 1/2$, but $F'(0; (1, 1)) = 2$. A linear derivative with the two axis values would instead give 1 on $(1, 1)$, a contradiction. The map is nonlinear and not Fréchet differentiable at the origin, despite exact agreement with F_0 on the coordinate axes. "Axis-calibrated linear counterpart" is therefore the appropriate term.

B.7.4 Where the central interaction comes from. As an analytic counterfactual, replace min rule weights by $w_{ab}^{\text{prod}} = \varphi_a(X)\varphi_b(Y)$, retaining the membership functions and singletons. On $|X|, |Y| \le 5$, the active consequents add their two single-input values and the denominator is $(\sum_a \varphi_a(X))(\sum_b \varphi_b(Y)) = 1$. Thus

$$F_{\text{prod}}(X, Y) = X/2 + Y/2 = F_0(X, Y). \tag{69}$$

The central nonlinear interaction is tied to normalized min activation, not merely to linguistic labels. The identity isolates the role of min activation; it does not assert linearity beyond the central square or an experimental comparison of product and min inference.

B.8 Nonlinear memory and what it does not identify

For identical scaled input pairs (X_k, Y_k), including consistent previous-error initialization, let A_k^{F} use F and A_k^0 use F_0, with equal initial accumulators. Subtracting their updates gives

$$A_k^{\text{F}} - A_k^0 = (A_{k-1}^{\text{F}} - A_{k-1}^0) + K_u \mathcal{R}_F(X_k, Y_k).$$

Telescoping yields

$$A_k^{\text{F}} - A_k^0 = K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{70}$$

With zero PI/Fuzzy constructor states and the calibrated parameters, the same-input bookkeeping can also be written

$$A_k^{\text{F}} = k_p E_k + \kappa_I \sum_{\ell=0}^{k} E_\ell + K_u \sum_{\ell=0}^{k} \mathcal{R}_F(X_\ell, Y_\ell). \tag{71}$$

The first two terms form a PI-like shadow evaluated on the Fuzzy trajectory's own inputs. The last term records accumulated nonlinear correction. If later inputs lie on the axes, the new residual is zero but the accumulated difference can remain nonzero.

This identity requires no accumulator projection/reset. Pure external clipping with retained raw states leaves the raw identity intact, but the applied difference must be evaluated through the limiter. Tracking/reset requires explicit additional state-change terms. Moreover, independently evolving PI and Fuzzy closed loops generate different measurements and possibly different physical event schedules. A same-input shadow identity therefore cannot be relabelled as their observed closed-loop difference.

B.9 Interpretation and the bridge to the stability section

The derivation provides implementation-consistent PI calibration, the exact central nonlinear map, and a regional sector suitable for a closed-loop certificate. Section C uses these properties after stating its plant, sampling, latency and initialization assumptions and proving that the sector's input region remains invariant. The present algebra does not change the continuous-clock Callisto plant into a discrete-time plant; it describes the controller updates embedded in that execution.

Same-sign scaled inputs indicate alignment of the two input channels. They do not always mean increasing absolute error: a transition from $e_{k-1} = -10$ to $e_k = 1$ has $e_k(e_k - e_{k-1}) > 0$ even though the absolute error decreased. Likewise, opposing-sign attenuation concerns a single increment relative to F_0, not an established network jitter attenuation property. Inter-node frequency differences, buffer excursion, common frequency and control activity must be evaluated separately.

C Stability, Safety and Certified Comparison: Full Proofs

Correspondence: Section 4. This appendix preserves the extended model, intermediate steps and supporting interpretation behind the corresponding main-text section.

We connect the nonlinear controller of Section B to four separate properties: invariance of its analyzed input region, convergence of inter-node frequency differences and buffer deviations, strict buffer-capacity safety, and positive physical clock frequency. We then establish a strict headroom comparison with its calibrated PI baseline for one specified initialization. The proof concerns an explicitly defined continuous fluid reference. Callisto retains asynchronous local sampling, delivery delays and integer frame events, which are not silently removed from its experimental model.

C.1 Reference assumptions and continuous-time model

C.1.1 What the reference retains and idealizes. The reference retains the controller of Section B, continuous clock phases, local occupancy feedback, the implemented PI integrate-before-output convention and the proposed incremental normalized-min Fuzzy surface. It assumes zero physical link latency, zero control-execution delay, fluid rather than integer occupancy and a common physical sampling grid $t_k = kH$, $H > 0$. A zero-order hold (ZOH) means that the applied frequency correction is constant on each interval $[t_k, t_{k+1})$. These are explicit analysis assumptions; local-clock polling in Callisto is not asserted to have this common grid. The equations sampled at kH below are exact endpoint relations for that continuous ZOH flow. They are neither numerical Euler integration nor a discrete firing-count replacement of the plant.

Let $\mathcal{N}_i^-$ be the fixed incoming-neighbor set and $d_i = |\mathcal{N}_i^-|$. The graph is connected, undirected and unit-weight, with a receiving channel in both directions. Its symmetric Laplacian L has diagonal d_i and off-diagonal -1 at neighboring nodes. Each undirected connection contributes once to this Laplacian; the pair of directed receiving channels does not double its weights.

The physical phase is θ_i, the uncontrolled frequency is ω_i^u, the applied correction is c_i, and the requested correction is q_i. The target occupancy of receiving edge $j \to i$ is $\beta^*_{j\to i}$. The fixed coefficient scale in the PI baseline law is $h = p/\omega_{\text{base}}$, where p is a local polling period and ω_{base} is the implementation reference frequency; H is the physical sampling period imposed only in this reference. The integral coefficient is $\kappa_I = k_i h$. These two time scales are kept distinct. All receiving buffers have common capacity $\beta_{\max} > 0$, and the uncontrolled frequencies ω_i^u are constant. The all-ones column vector is $\mathbf{1}$.

The transpose is denoted by $\top$. A symmetric matrix $A \succeq 0$ is positive semidefinite; $A \succ 0$ is positive definite. The matrix order $A \preceq B$ means $B - A \succeq 0$. The symbol $\otimes$ denotes the Kronecker product, and $\|z\|_\infty = \max_i |z_i|$.

C.1.2 Compatible occupancy targets and marking. Throughout the proved centered reference, all initial phases are equal. For zero delay the fluid law in the additive phase-offset convention is

$$\beta_{j\to i}(t) = \theta_j(t) - \theta_i(t) + \lambda_{j\to i}. \tag{72}$$

Here $\lambda_{j\to i}$ is the fixed additive phase offset. With equal initial phases it also equals the initial marking; this presentation does not identify these conventions for arbitrary unequal initial phases. The law follows the bittide fluid modeling framework [1, 3]. Define $\beta^*_{j\to i} = \theta_j(0) - \theta_i(0) + \lambda_{j\to i}$, so the reference starts at its compatible targets. For a common constant ramp frequency ω_{ref} put

$$x_k = \theta(t_k) - \theta(0) - t_k \omega_{\text{ref}} \mathbf{1}. \tag{73}$$

The common ramp cancels from edge differences. Subtracting the target in (72) gives

$$\beta_{j\to i}(t_k) - \beta^*_{j\to i} = x_{j,k} - x_{i,k}, \qquad E_k = -Lx_k. \tag{74}$$

The vector E collects the Section B frame errors, actual minus target. Not every set of per-edge targets is compatible with a fixed marking and phase initialization; that compatibility is an assumption of this model.

For the worked example, equal initial phases and target 1024 imply $\lambda_{j\to i} = 1024$. The delayed Callisto simulations have marking 1044 with 20 initial in-flight frames; simply retaining 1044 while imposing zero delay and the same centered occupancy would change the initialization.

C.2 From the original controller to the sampled recurrence

Define the uncontrolled frequency offsets $\delta\omega = \omega^u - \omega_{\text{ref}}\mathbf{1}$ and the physical frequency offsets $w_k = \delta\omega + c_k$. For $k \geq 1$, let $v_k = Hw_{k-1}$. The plant's previous genuine interval gives

$$x_k - x_{k-1} = Hw_{k-1} = v_k. \tag{75}$$

Indeed, on each zero-order-held interval,

$$x(t) = x_k + (t - t_k)w_k, \qquad t \in [t_k, t_{k+1}], \tag{76}$$

where $x(t) = \theta(t) - \theta(0) - t\omega_{\text{ref}}\mathbf{1}$. Integration gives $x_{k+1} = x_k + Hw_k$ exactly. The controller is updated only at the endpoints, but the plant evolves continuously throughout the interval.

The factor H is physical elapsed time; the controller factor h remains the fixed PI coefficient scale. Section B uses $e_i = -\gamma_i E_i$, so the Fuzzy inputs are

$$X_k = K_e \Gamma L x_k, \qquad Y_k = K_{\Delta e} \Gamma L v_k, \qquad \Gamma = \operatorname{diag}(\gamma_1, \ldots, \gamma_n). \tag{77}$$

Here $K_e, K_{\Delta e}$ are chosen common for readability; node-dependent output scaling can still produce common calibrated k_p, k_i.

On $\Omega_r = \{(X, Y) : |X| + |Y| \leq r\}$, $0 < r \leq 5$, Section B gives $F = \alpha F_0$, where $F_0 = (X + Y)/2$ and $\underline{\alpha} = (1 + r/5)^{-1} \leq \alpha \leq \overline{\alpha} = 2$. With $D_{\alpha,k} = \operatorname{diag}(\alpha_{1,k}, \ldots, \alpha_{n,k})$, the calibrated increment becomes

$$c_k - c_{k-1} = -D_{\alpha,k}(\kappa_I L x_k + k_p L v_k). \tag{78}$$

Multiplication by H and the next plant interval yield

$$\begin{aligned} v_{k+1} &= v_k - D_{\alpha,k}(bLx_k + aLv_k), \\ x_{k+1} &= x_k + v_{k+1}, \qquad a = Hk_p, \quad b = H\kappa_I = Hk_i h. \end{aligned} \tag{79}$$

The extra H in b is essential: v is a scaled frequency. The surface factor $\alpha_{i,k}$ is node-dependent and state-dependent; freezing it and checking one constant-gain PI characteristic is insufficient.

C.2.1 The initial constructor call. The controller's stored previous error is not a previous plant interval. For the centered initialization $x_0 = 0$ with zero PI sample sum, zero Fuzzy accumulator and zero previous percentage error, the first original controller call gives $c_0 = 0$. Advancing $[0, H]$ then gives

$$x_1 = H\delta\omega, \qquad v_1 = H\delta\omega. \tag{80}$$

Only from $k = 1$ does (75) justify the recurrence. Nonzero initial occupancy error is outside this centered formulation: one must also introduce target-compatible phase coordinates or retain the affine error bias, then explicitly evaluate the original constructor and recheck the initial state and certificate. Evaluating the first call alone does not restore $E = -Lx$. The centered bootstrap is consistent with, but separate from, the event-level first-call PI/F_0 equivalence proved in Section B.

C.3 Disagreement projection and the symmetric sector matrix

Since $L\mathbf{1} = 0$, a common phase shift does not change occupancies. Choose U with orthonormal columns spanning the subspace orthogonal to $\mathbf{1}$, so $U^\top U = I_{n-1}$ and $U^\top \mathbf{1} = 0$. Choose its columns as graph eigenvectors, giving

$$L = U\Lambda U^\top, \qquad \Lambda = \operatorname{diag}(\mu_2, \dots, \mu_n), \quad 0 < \mu_2 \le \mu_n. \tag{81}$$

I_{n-1} denotes the identity matrix of dimension $n-1$. Define

$$\xi_k = \Lambda^{1/2} U^\top x_k, \qquad \eta_k = \Lambda^{1/2} U^\top v_k, \qquad y_k = (\xi_k^\top, \eta_k^\top)^\top. \tag{82}$$

The symbol ξ is projected phase and η already contains the frequency scale H. The coordinates retain the physical-time scale H; in particular, η is a scaled frequency disagreement rather than an integral-controller state.

For $M_k = \Lambda^{1/2} U^\top D_{\alpha,k} U \Lambda^{1/2}$, substitution in (79) gives

$$\begin{aligned} \eta_{k+1} &= -bM_k\xi_k + (I_{n-1} - aM_k)\eta_k, \\ \xi_{k+1} &= (I_{n-1} - bM_k)\xi_k + (I_{n-1} - aM_k)\eta_k. \end{aligned} \tag{83}$$

In particular, the diagonal D_α need not commute with L; the projection produces the symmetric M_k instead of assuming shared modes. Its quadratic form satisfies

$$\underline{\alpha}\Lambda \preceq M_k \preceq \overline{\alpha}\Lambda, \qquad \operatorname{eig}(M_k) \subseteq [\nu_-, \nu_+] = [\underline{\alpha}\mu_2, \overline{\alpha}\mu_n]. \tag{84}$$

The scalar ν used below is a possible eigenvalue of M, not a node frequency or the number of communication edges.

C.4 A common Lyapunov certificate from two endpoints

For matrices $M \in \mathbb{R}^{(n-1)\times(n-1)}$ and scalars ν, define

$$\mathcal{B}(M) = \begin{bmatrix} I - bM & I - aM \\ -bM & I - aM \end{bmatrix}. \tag{85}$$

The identity blocks have dimension $n-1$. Equations above give $y_{k+1} = \mathcal{B}(M_k) y_k$. A scalar block associated with an eigenvalue of M_k is

$$B(\nu) = \begin{bmatrix} 1 - b\nu & 1 - a\nu \\ -b\nu & 1 - a\nu \end{bmatrix}. \tag{86}$$

For a symmetric positive-definite matrix $\bar{P} \in \mathbb{R}^{2\times 2}$, put $P = \bar{P} \otimes I_{n-1}$. The scalar $V_k = y_k^\top P y_k$ is a quadratic measure of phase and frequency disagreement, not of absolute phase.

Lemma C.1 (Endpoint certificate). *Suppose* $0 < \rho < 1$ *and*

$$\begin{aligned} B(\nu_-)^\top \bar{P} B(\nu_-) &\preceq \rho \bar{P}, \\ B(\nu_+)^\top \bar{P} B(\nu_+) &\preceq \rho \bar{P}. \end{aligned} \tag{87}$$

Then every update whose inputs satisfy the regional sector obeys

$$V_{k+1} \le \rho V_k. \tag{88}$$

The conclusion permits node-dependent and time-varying sector factors.

Proof. First consider one possible scalar eigenvalue. For $\nu = (1-t)\nu_- + t\nu_+$, $0 \le t \le 1$, affine dependence gives $B(\nu) = (1-t)B_- + tB_+$, where $B_\pm = B(\nu_\pm)$. Expanding the two quadratic expressions yields

$$\begin{aligned} &(1-t)B_-^\top \bar{P} B_- + tB_+^\top \bar{P} B_+ - B(\nu)^\top \bar{P} B(\nu) \\ &\quad = t(1-t)(B_+ - B_-)^\top \bar{P} (B_+ - B_-) \succeq 0. \end{aligned} \tag{89}$$

Together with the endpoint assumptions, this proves $B(\nu)^\top \bar{P} B(\nu) \preceq \rho\bar{P}$ throughout the continuous interval $[\nu_-, \nu_+]$. No eigenvalue grid is required.

Next, each M_k is real symmetric, so it admits an orthogonal factorization $M_k = Q_k \operatorname{diag}(\nu_{1,k}, \dots, \nu_{n-1,k}) Q_k^\top$. Rotating both ξ_k and η_k by $Q_k^\top$ reduces the dynamics to the scalar blocks $B(\nu_{\ell,k})$. The corresponding block rotation leaves $P = \bar{P} \otimes I_{n-1}$ unchanged. Thus summing the scalar inequalities gives $\mathcal{B}(M_k)^\top P \mathcal{B}(M_k) \preceq \rho P$, which is (88). Because the witness P is the same for every permitted M_k, Q_k may change at every step. No simultaneous diagonalization of the node-dependent matrices is assumed. □

The remaining task is to show that the actual nonlinear inputs stay in the region where the sector was obtained. Stability of a family of hypothetical linear blocks alone would not establish that fact.

C.5 Closing the invariant input region

Let $\mathbf{e}_i$ denote the ith unit vector and define the row $a_i^\top = \mathbf{e}_i^\top U \Lambda^{1/2}$. Its squared Euclidean norm is $\|a_i\|^2 = L_{ii} = d_i$. The Fuzzy inputs become

$$X_{i,k} = \gamma_i K_e a_i^\top \xi_k, \qquad Y_{i,k} = \gamma_i K_{\Delta e} a_i^\top \eta_k. \tag{90}$$

For any vector c and $P \succ 0$, Cauchy–Schwarz in the quadratic metric gives $(c^\top y)^2 \le (y^\top P y)(c^\top P^{-1} c)$. Also $|X| + |Y| = \max_{\sigma=\pm 1} |X + \sigma Y|$. Applying that output bound on $V \le R$ yields

$$\begin{aligned} |X_{i,k}| + |Y_{i,k}| &\le \sqrt{R\Psi_i}, \\ \Psi_i &= \gamma_i^2 d_i \max_{\sigma=\pm 1} [K_e, \sigma K_{\Delta e}] \bar{P}^{-1} [K_e, \sigma K_{\Delta e}]^\top. \end{aligned} \tag{91}$$

The scalar Ψ_i is the largest squared input gain on the energy ellipsoid. If $V_1 \le R$ and $R\Psi_i \le r^2$ for every node, then: each node's first scaled input pair belongs to Ω_r; the valid sector gives $V_2 \le \rho V_1 \le R$; the same output bound keeps the next inputs in Ω_r; induction repeats. This proves the sector domain and disagreement ellipsoid invariant together, rather than assuming either.

Equivalently, a valid radius must satisfy

$$V_1 \le R \le \min_i \frac{r^2}{\Psi_i}. \tag{92}$$

The initialization, normalization through d_i and $\beta_{\max}$, the Fuzzy input scales and the graph-dependent witness therefore enter the same condition. When $\Psi_i = 0$, that node places no restriction on R; otherwise the displayed quotient is finite. The inference surface is still nonlinear inside the invariant region. The sector is an enclosure of its instantaneous response, not an assertion that its gain is fixed.

Because $P \succ 0$ and $V_k \le \rho^{k-1} V_1$, the projected phase and frequency decay exponentially. The ellipsoid constrains disagreement; the common physical phase continues to grow and is not bounded by V. Thus convergence means agreement of rates and restoration of compatible occupancy targets, not convergence of the absolute clock phases to constants.

C.6 From disagreement to physical buffer and clock bounds

C.6.1 Edge occupancy and intersample time. For edge $e = (j \to i)$, define its signed unit incidence vector $\mathbf{b}_e = \mathbf{e}_j - \mathbf{e}_i$. Since it is

orthogonal to $\mathbf{1}$, its output from the projected phase is

$$\mathbf{b}_e^\top x_k = \mathbf{b}_e^\top U\Lambda^{-1/2}\xi_k.$$

Define the graph pseudoinverse $L^\dagger = U\Lambda^{-1}U^\top$ and effective resistance $R_e^{\mathrm{eff}} = \mathbf{b}_e^\top L^\dagger \mathbf{b}_e$. The quadratic output inequality gives

$$|\beta_e(t_k) - \beta_e^*|^2 \le V_k[\bar{P}^{-1}]_{11}R_e^{\mathrm{eff}}. \tag{93}$$

Under ZOH the phase, hence each fluid occupancy, is affine on an interval. For $0 \le s \le 1$ its deviation is a convex combination of endpoint deviations, so its absolute value cannot exceed the larger endpoint absolute value. This covers all physical time, including the initial interval from the zero state to x_1. It would not apply unchanged to arbitrary delayed integer arrival events.

Corollary C.2 (Buffer safety within the reference model). *Suppose the compatible centered initialization and the invariant-domain conditions above hold with $V_1 \le R$. For each receiving edge define*

$$m_e = \min\{\beta_e^*, \beta_{\max} - \beta_e^*\},$$
$$B_e(R) = \sqrt{R[\bar{P}^{-1}]_{11}R_e^{\mathrm{eff}}}. \tag{94}$$

If $0 < \beta_e^* < \beta_{\max}$ and $B_e(R) < m_e$ on every receiving edge, then the unconstrained fluid execution satisfies $0 < \beta_e(t) < \beta_{\max}$ for all $t \ge 0$. In addition, $\beta_e(t) - \beta_e^* \to 0$ as $t \to \infty$.

Proof. At t_k, the squared deviation is bounded by $V_k[\bar{P}^{-1}]_{11}R_e^{\mathrm{eff}}$ and hence by $B_e(R)^2$. The initial deviation is zero. On $[t_k, t_{k+1}]$, the deviation is a convex combination of its endpoint deviations, so its absolute value is at most $B_e(R)$. The strict headroom inequalities give the lower and upper capacity inequalities. Finally $V_k \le \rho^{k-1}V_1 \to 0$ and the same interpolation argument imply convergence at every physical time, not only at samples. □

Here $B_e(R) < m_e$ is a sufficient condition. If it fails, the certificate is inconclusive; failure of this conservative test does not prove a buffer overflow or underflow. The condition also concerns the unconstrained flow. It does not identify a protection-induced stall with successful control convergence.

C.6.2 Frequency-difference safety and convergence. Define the physical frequency spread by

$$\Delta_\omega(t) = \max_i \omega_i(t) - \min_i \omega_i(t). \tag{95}$$

A small frequency spread means close clock frequencies; it does not require the phases or their integer counts to coincide. For any pair $i \ne j$, not necessarily adjacent, define

$$R_{ij}^{\mathrm{eff}} = (\mathbf{e}_i - \mathbf{e}_j)^\top L^\dagger(\mathbf{e}_i - \mathbf{e}_j), \qquad R_{\max}^{\mathrm{eff}} = \max_{i\ne j} R_{ij}^{\mathrm{eff}}, \tag{96}$$

where $\mathbf{e}_i$ is the ith unit vector. Connectedness makes these quantities finite. During $[t_k, t_{k+1})$, $\omega_i(t) = \omega_{\mathrm{ref}} + w_{i,k}$, while $v_{k+1} = Hw_k$. Therefore

$$\begin{aligned}|\omega_i(t) - \omega_j(t)|^2 &= H^{-2}|(\mathbf{e}_i - \mathbf{e}_j)^\top v_{k+1}|^2\\ &\le H^{-2}V_{k+1}[\bar{P}^{-1}]_{22}R_{ij}^{\mathrm{eff}}\\ &\le H^{-2}R\rho^k[\bar{P}^{-1}]_{22}R_{ij}^{\mathrm{eff}}.\end{aligned} \tag{97}$$

The factor V_{k+1}, rather than V_k, is essential: v_k represents the frequency held during the previous interval. The case $k = 0$ is covered by the explicitly constructed bootstrap state y_1. At switching times the same bound applies to the newly held frequency; the previous interval bounds the left-hand limit.

Corollary C.3 (A prescribed frequency-spread threshold). *Under the invariant-domain conditions, let $\Delta_{\mathrm{safe}} > 0$ be a specified engineering tolerance on pairwise frequency differences. If*

$$D(R) := \frac{1}{H}\sqrt{R[\bar{P}^{-1}]_{22}R_{\max}^{\mathrm{eff}}} < \Delta_{\mathrm{safe}}, \tag{98}$$

then for every $t \in [t_k, t_{k+1})$, $k \ge 0$,

$$\Delta_\omega(t) \le D(R)\rho^{k/2} < \Delta_{\mathrm{safe}}, \qquad \Delta_\omega(t) \longrightarrow 0. \tag{99}$$

Proof. The spread equals $\max_{i,j}|\omega_i - \omega_j|$. Maximizing (97) and taking square roots proves the bound. Since $0 < \rho < 1$, the bound tends to zero. □

The threshold is a user-specified performance or hardware tolerance; logical synchrony does not by itself prescribe its numerical value. This proof bounds the transient as well as the limiting disagreement. It neither proves a smaller Fuzzy frequency peak than PI nor adds a guarantee for arbitrary delayed or stochastic executions. Frequency spread also cannot certify positive absolute frequency: a collection of equal nonpositive rates would have zero spread. The common frequency must therefore be bounded separately.

C.6.3 The unprojected common frequency. Let $\bar{w}_k = n^{-1}\mathbf{1}^\top w_k$ be the common-frequency offset. Put $\alpha_c = (\overline{\alpha}+\underline{\alpha})/2$ and $s_\alpha = (\overline{\alpha}-\underline{\alpha})/2$. With $g = (\kappa_I, k_p)^\top$, averaging (78) and using $\mathbf{1}^\top U = 0$ gives

$$\bar{w}_k - \bar{w}_{k-1} = -\frac{1}{n}\mathbf{1}^\top(D_{\alpha,k} - \alpha_c I_n)U\Lambda^{1/2}(\kappa_I\xi_k + k_p\eta_k). \tag{100}$$

The second component of g is k_p, not Hk_p, since η already contains H. Bounding each factor yields

$$|\bar{w}_k - \bar{w}_{k-1}| \le s_\alpha\sqrt{\frac{\mu_n}{n}g^\top\bar{P}^{-1}g}\sqrt{V_k}. \tag{101}$$

The geometric series in $\sqrt{\rho}$ is summable. Hence

$$\begin{aligned}\sup_k|\bar{w}_k| &\le |\bar{w}_0| + \frac{s_\alpha\sqrt{(\mu_n/n)Rg^\top\bar{P}^{-1}g}}{1-\sqrt{\rho}}\\ &\le |\bar{w}_0| + \frac{2s_\alpha}{1-\rho}\sqrt{\frac{\mu_n}{n}Rg^\top\bar{P}^{-1}g} = W_c.\end{aligned} \tag{102}$$

The last inequality follows from $(1-\sqrt{\rho})^{-1} = (1+\sqrt{\rho})/(1-\rho) \le 2/(1-\rho)$. Absolute summability also proves convergence of $\bar{w}_k$. Combined with vanishing disagreement, all node frequencies converge to one value $\omega_\infty = \omega_{\mathrm{ref}} + \lim_k \bar{w}_k$. For Fuzzy PI that limit need not be the mean initial uncontrolled frequency.

C.6.4 Closing actuator saturation and positive frequency. The ith disagreement-frequency output satisfies

$$|w_{i,k} - \bar{w}_k| \le W_d, \qquad W_d = \max_i \frac{1}{H}\sqrt{R[\bar{P}^{-1}]_{22}(L^\dagger)_{ii}}. \tag{103}$$

Applying the state bound to y_{k+1} covers w_k; y_1 covers w_0. As $c = w - \delta\omega$, we obtain

$$|c_{i,k}| \le W_c + W_d + \|\delta\omega\|_\infty, \qquad \omega_i(t) \ge \omega_{\mathrm{ref}} - W_c - W_d. \tag{104}$$

First derive these bounds for the unconstrained trajectory. If the first is strictly below $c_{\max}$ and the second is positive, the limiter is never reached, so the constrained execution coincides with that trajectory. This continuation argument closes the assumption of

no clipping. The buffer bound must likewise fit within physical capacity; protecting a buffer by blocking execution is a different outcome.

Together, the following sufficient conditions connect the controller and network parameters to the required properties:

$$\begin{aligned}
&V_1 \le R, \qquad R\Psi_i \le r^2 \quad \text{for every } i,\\
&0 < \beta_e^* < \beta_{\max} \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{11} R_e^{\text{eff}} < m_e^2 \quad \text{for every } e,\\
&R[\bar{P}^{-1}]_{22} R_{\max}^{\text{eff}} < H^2 \Delta_{\text{safe}}^2,\\
&W_c + W_d + \|\delta\omega\|_\infty < c_{\max},\\
&\omega_{\text{ref}} - W_c - W_d > 0,
\end{aligned} \tag{105}$$

provided the endpoint inequalities of Lemma C.1 hold. The frequency-spread inequality is required only when a specific threshold is an objective; convergence alone does not require choosing that threshold. All quantities are evaluated for one consistent model, initialization and parameter set. In particular, changing H, h, $\beta_{\max}$, node degree, controller gains or topology can change more than one inequality. These are sufficient certification tests, not a characterization of the largest safe set or necessary stability boundaries.

C.7 The prescribed triangle certificate

Take the three-node complete graph, giving $\mu_2 = \mu_3 = 3$, $R_e^{\text{eff}} = 2/3$ and $(L^\dagger)_{ii} = 2/9$. The parameters are

$$\begin{aligned}
&H = h = \frac{30000000}{29999}, \quad k_p = \frac{1}{500000}, \quad k_i = \frac{1}{20000000000},\\
&K_e = \frac{1}{10}, \quad K_{\Delta e} = \frac{29999}{7500}, \quad K_u = -\frac{768}{18749375},\\
&\beta_{\max} = 2048, \quad \beta_e^* = \beta_e(0) = 1024, \quad \lambda_e = 1024.
\end{aligned} \tag{106}$$

Frequencies in this numerical example are normalized, and physical time is measured in reference ticks. Use the uncontrolled frequency vector

$$\omega^u = (1.0002, 0.9998, 0.9999)^\top$$

and its average as $\omega_{\text{ref}} = 29999/30000$, so $\delta\omega = (7, -5, -2)^\top/30000$ and $\bar{w}_0 = 0$. At the compatible startup (80), use

$$\bar{P} = \frac{1}{50000}\begin{bmatrix} 9 & 181 \\ 181 & 49629 \end{bmatrix}, \quad \rho = \frac{1599}{1600}, \quad r = 1. \tag{107}$$

Here $\det \bar{P} = 4139/25000000 > 0$ and the leading diagonal entry is positive. Since the sum of its four entries is one,

$$R_0 = V_1 = H^2 \delta\omega^\top L \delta\omega = \frac{234000000}{899940001}. \tag{108}$$

This exact fraction is used in all inequalities. The displayed approximation 0.2600173342 is slightly below the exact value and is not substituted as an upper energy radius.

The sector is $[5/6, 2]$, hence the spectral endpoints are $5/2$ and 6. For $D_\nu = \rho\bar{P} - B(\nu)^\top \bar{P} B(\nu)$, exact rational arithmetic verifies positive leading minor and determinant at both ends:

ν	$[D_\nu]_{11}$ (approx.)	$\det D_\nu$ (approx.)
5/2	$8.2193625299 \times 10^{-7}$	$7.8876135318 \times 10^{-11}$
6	$2.0776400066 \times 10^{-6}$	$7.3380136479 \times 10^{-9}$

The accompanying machine-readable certificate stores the exact fractions; the decimals are only a readable display, not rounded tests of positivity. Using R_0 gives the following conservative, outward bounds:

Quantity	Certified bound
$\sup_{i,k}(\lvert X_{i,k}\rvert + \lvert Y_{i,k}\rvert)$	$< 0.171455 < 1$
Coarse edge excursion	< 32.237442 frames
$\sup_k \lvert\bar{w}_k\rvert$	< 0.003682181
$\sup_{i,k} \lvert w_{i,k} - \bar{w}_k\rvert$	< 0.000250634
$\sup_{i,k} \lvert c_{i,k}\rvert$	$< 0.004166148 < 0.01$
$\inf_{i,t} \omega_i(t)$	> 0.99603385

The coarse excursion already fits within the 1024-frame lower and upper headroom. The frequency bounds use the exact rational-square-root enclosures, not a sum of prematurely rounded display values.

C.8 The exact PI all-time baseline

For $J \in \{\text{PI}, \text{F}\}$ define

$$B_J = \sup_{t \ge 0} \max_e |\beta_e^J(t) - 1024|. \tag{109}$$

The supremum varies time and receiving edges, while the initialization and uncontrolled-frequency vector remain fixed. For the same plant, initialization and gains, the PI baseline has $D_\alpha = I_n$. Averaging its increments shows that the mean frequency is preserved. On the triangle, both disagreement modes have eigenvalue 3. Define $\chi_0 = 0$, $\chi_1 = 1$ and

$$\chi_{k+1} = (2 - 3a - 3b)\chi_k - (1 - 3a)\chi_{k-1}. \tag{110}$$

To derive it, use $v_k = x_k - x_{k-1}$ in the PI state recurrence:

$$x_{k+1} = (2I_n - (a+b)L)x_k - (I_n - aL)x_{k-1}.$$

Since $L\delta\omega = 3\delta\omega$ and $x_1 = H\delta\omega$, induction gives $x_k = H\chi_k\delta\omega$. With $\Delta_0 = \max_i \omega_i^u - \min_i \omega_i^u = 1/2500$,

$$\max_e |\beta_e(t_k) - 1024| = H\Delta_0|\chi_k|. \tag{111}$$

This exact rational sequence is enumerated through $N = 200$. Its unique prefix maximum occurs at $k = 111$, with

$$23.40370130812701 \le \max_{0 \le k \le N} H\Delta_0|\chi_k| \le 23.40370130812702. \tag{112}$$

It is not yet an all-time assertion until the tail is bounded.

At $N = 200$ put $\delta\chi_N = \chi_N - \chi_{N-1}$, the scalar response difference. The exact state energy is

$$V_N = R_0\{\bar{P}_{11}\chi_N^2 + 2\bar{P}_{12}\chi_N\delta\chi_N + \bar{P}_{22}(\delta\chi_N)^2\}. \tag{113}$$

The common certificate also covers PI's scalar value $\nu = 3$. Thus for all $k \ge N$, the remaining buffer excursion is at most $\sqrt{(2/3)[\bar{P}^{-1}]_{11} V_N}$, with an outward upper bound less than 17.958057 frames. This is strictly below the attained prefix maximum. ZOH interpolation therefore makes (112) the exact all-physical-time B_{PI} enclosure, reached at time $111H$.

C.9 The Fuzzy interval prefix and contracting tail

A direct floating-point trajectory can locate a candidate peak but does not enclose the exact one. The certificate instead propagates

intervals for all components of x_k and v_k on a grid $S^{-1} = 10^{-60}$. A rational number q_{rat} is enclosed by

$$\left[\frac{\lfloor Sq_{\mathrm{rat}}\rfloor}{S}, \frac{\lceil Sq_{\mathrm{rat}}\rceil}{S}\right].$$

Each multiplication uses the extrema of the four endpoint products, rounded outward; division uses a strictly positive enclosed denominator and outward-rounded endpoint quotients. Addition and subtraction use endpoint arithmetic. These operations use integers and rational comparisons, not floating-point rounding assumptions.

For the central formula, $m = \min(|X|, |Y|)$ is enclosed by interval absolute values and endpoint minima. Its interaction term can be written $J = \min(X, Y)$ for two positive inputs, $J = \max(X, Y)$ for two negative inputs and zero otherwise. It is coordinatewise nondecreasing, including across the axes; its lower and upper corner values enclose it on any input rectangle. The positive denominator $1 + 2m/5$ then permits a valid interval evaluation of the exact map. The invariant-domain proof ensures that this central formula is applicable to the exact trajectory.

Propagating the genuine bootstrap state through state $N = 200$ gives a prefix maximum enclosed by

$$[17.314056817962,\ 17.314056817963] \quad \text{frames.} \tag{114}$$

The prefix retains all node coordinates, so common drift is not silently discarded. On the triangle the quadratic energy is equivalently

$$\begin{aligned} V_k = \sum_{i<j} \Big\{ & \bar{P}_{11}(x_{i,k} - x_{j,k})^2 \\ & + 2\bar{P}_{12}(x_{i,k} - x_{j,k})(v_{i,k} - v_{j,k}) \\ & + \bar{P}_{22}(v_{i,k} - v_{j,k})^2 \Big\}. \end{aligned} \tag{115}$$

Evaluating this expression outward gives $\overline{V}_N \geq V_N$. For all later samples,

$$|\beta_e(t_k) - 1024|^2 \leq \frac{2}{3}[\bar{P}^{-1}]_{11}\rho^{k-N}\overline{V}_N. \tag{116}$$

For either controller, an equivalent tail bound that makes the finite-prefix/infinite-tail decomposition explicit is

$$T_N = \sqrt{\frac{2}{3}[\bar{P}^{-1}]_{11}\overline{V}_N}. \tag{117}$$

For PI one can take $\overline{V}_N = V_N$ exactly; for Fuzzy, $\overline{V}_N$ is an outward interval upper bound. Once T_N is below the certified prefix lower bound, no later sample can exceed the prefix maximum. This closes the infinite time horizon without selecting an arbitrarily long simulation duration.

An outward upper bound for this whole tail is 12.29271005372874 frames, strictly below the prefix lower bound. Including $x_0 = 0$ and intersample interpolation gives $B_{\mathrm{F}} < 17.315$ on the entire physical time axis. The exact Fuzzy peak is more tightly enclosed by (114); the rounded 17.315 value is used for a simple, conservative comparison.

The enclosures used in the comparison are collected in Table 6. The prefix intervals are outward enclosures of actual peak values; the tail entries are upper bounds. Their different roles are essential for the strict comparison.

Table 6: Certified excursions in the prescribed fluid reference (frames).

Controller	Prefix peak enclosure	Tail upper bound
PI	[23.40370130812701, 23.40370130812702]	17.95805698665804
Fuzzy	[17.314056817962, 17.314056817963]	12.29271005372874

C.10 Strict comparison, engineering meaning and scope

Theorem C.4 (Strict headroom benefit for a prescribed initialization). *For the triangle, parameters and compatible initialization above, initialize the PI sample sum, the Fuzzy accumulator and the stored previous percentage error to zero. Both reference executions keep positive physical frequencies, remain strictly inside the buffer interval* $(0, 2048)$ *and never activate the* 0.01 *correction limiter. Buffer deviations and all pairwise frequency differences converge to zero. Their all-time excursions satisfy*

$$B_{\mathrm{F}} < 17.315 < 23.40370130812701 \leq B_{\mathrm{PI}}. \tag{118}$$

For the minimum lower/upper headroom $M_J = 1024 - B_J$,

$$M_{\mathrm{F}} > 1006.685, \qquad M_{\mathrm{F}} - M_{\mathrm{PI}} > 6.08870130812701 \quad \textit{frames}. \tag{119}$$

Proof. The endpoint inequalities and the radius test certify the invariant nonlinear input region. The buffer, correction and absolute-frequency bounds establish capacity safety, absence of clipping and positive frequencies. The invariant disagreement energy contracts, so the buffer and frequency-difference conclusions follow. PI is included by taking $\alpha_i = 1$, which belongs to the same sector; PI also preserves its initial mean frequency. The certified finite-prefix enclosures and the tail bounds imply that both all-time excursions are attained in the prefix. The Fuzzy prefix upper bound is below 17.315, whereas PI attains the enclosed peak at sample 111. Intersample affine occupancy rules out a missed larger fluid peak. Finally, each edge has a reverse edge with the negative occupancy deviation, so the worst lower/upper headroom is exactly $1024 - B_J$. Subtracting the strict excursion bounds gives the stated gain in headroom. □

The comparison uses a certified Fuzzy upper bound and an attained PI peak. Merely comparing two conservative upper bounds would not establish the strict ordering of the actual excursions.

At the original target 1024, both reserves are already large. That operating point alone does not demonstrate a storage bottleneck or hardware cost reduction. The following compatible-target family makes the safety significance of the strict excursion ordering explicit.

The supremum is over time and edges for one prescribed uncontrolled-frequency vector. It is not an expanded safe set for all 400-ppm patterns, a guarantee against arbitrary latency perturbations or superiority over independently tuned PI. For a different capacity, $\gamma_i = 100/(d_i\beta_{\max})$ changes; keeping the Fuzzy gains unchanged generally changes the frame-count nonlinear map. A new capacity certificate requires a fresh derivation or a

verified map-preserving normalization change. Physical occupancy constraints also do not replace the delayed runtime's in-flight and credit/readiness rules.

This is a computer-assisted mathematical certificate with reproducible integer/rational checks. It is not proof-assistant verification of the Julia implementation. The actual asynchronous, delayed, quantized cases belong in the experimental section. This division preserves the distinction between a theorem about the declared reference and empirical evidence about Callisto.

C.11 From excursion bounds to safe occupancy configurations

The reference theorem can express a configuration-domain advantage without changing capacity or the frame-count nonlinear map. Let J denote PI or Fuzzy and consider its unprotected fluid extension. Keep the graph, $\beta_{\max}$, gains, normalization, sampling, uncontrolled frequencies, equal initial phases and zero controller constructors fixed. For a scalar $z \in (0, \beta_{\max})$, set

$$\beta_{ji}(0) = \beta^*_{ji} = \lambda_{ji} = z \quad \text{on every directed receiving channel.} \tag{120}$$

This is a family of compatible initial/target configurations. The marking changes with z; the family does not describe arbitrary initial occupancy error at a fixed target or fixed application marking. PI and Fuzzy use the same configuration for each comparison.

Lemma C.5 (Translation invariance at fixed capacity). *For each J, its unprotected phase, requested/applied correction, controller memory and frequency trajectories are independent of z. In particular,*

$$\beta^{J,z}_{ji}(t) = \beta^{J,1024}_{ji}(t) + z - 1024. \tag{121}$$

Proof. Equal initial phases and zero-delay markings give $\beta^{J,z}_{ji} = z + \theta^J_j - \theta^J_i$. At initialization, all measured errors are zero, as are the stored constructors, for every z. Suppose that the phase and memory states agree through one update. Then

$$E^{J,z}_i = \sum_{j \in \mathcal{N}_i^-} (\beta^{J,z}_{ji} - z) = \sum_{j \in \mathcal{N}_i^-} (\theta^J_j - \theta^J_i)$$

does not depend on z. Its percentage form $e_i = -100E_i/(d_i\beta_{\max})$ is also unchanged because capacity and degree are fixed. Both the PI sample sum and the Fuzzy stored error and accumulator therefore update identically within their respective controller families. An identical external limiter preserves equality of applied corrections, and exact continuous held flow preserves the phase states to the next update. Induction establishes the identity, including every intersample time. If a guarded model is considered, the argument applies only up to its first guard intervention; the subsequent guarded trajectory requires separate analysis. □

For a bidirectional graph, reverse channels have opposite phase deviations. For the fixed frequency initialization define

$$B_J = \sup_{t \geq 0,\, j \to i} |\theta^J_j(t) - \theta^J_i(t)|.$$

The minimum two-sided headroom over time and channels is exactly

$$M_J(z) = \min(z, \beta_{\max} - z) - B_J. \tag{122}$$

Thus the targets with a strictly positive uniform reserve are $\mathcal{I}_J = (B_J, \beta_{\max} - B_J)$ when this interval is nonempty. The certified prefix/tail comparison proves $B_F < B_{\mathrm{PI}} < \beta_{\max}/2$, so $\mathcal{I}_{\mathrm{PI}} \subsetneq \mathcal{I}_F$ for this configuration family. In particular, $(17.315, 2048 - 17.315)$ is a sufficient Fuzzy target interval containing the entire PI safe target interval. At $z = 20$,

$$\inf_{e,t} \beta^{F,20}_e(t) > 2.685, \qquad \sup_{e,t} \beta^{F,20}_e(t) < 37.315.$$

At the attained PI prefix maximum $t_* = 111H$, a reverse channel has $\beta^{\mathrm{PI},20}_e(t_*) = 20 - B_{\mathrm{PI}} < -3.403701$. By continuity its unprotected occupancy reaches zero before t_*. This predicts a capacity violation of the unprotected reference, not a negative occupancy in a physically guarded implementation. Input invariance, positive frequency and inactive limiter remain certified because their phase/control trajectories are unchanged. The refined excursion bound supplies the capacity proof even where the coarser 32.237442-frame Lyapunov output bound is insufficient.

C.11.1 A directional initial-frequency capacity threshold. Keep the same mean frequency and let

$$\delta\omega(s) = \frac{s}{30000}(7, -5, -2)^\top, \quad 0 \leq s \leq 1.$$

The PI recurrence and its zero initial memory are linear in s. Its common frequency is conserved, and the deviations and corrections are scaled copies of the certified $s = 1$ trajectory. Positivity and absence of clipping therefore hold on this whole parameter interval. Consequently its exact excursion is $sB_{\mathrm{PI}}(1)$. At target $z = 20$, upper capacity is not restrictive before the lower boundary; the capacity threshold is

$$s_{\mathrm{PI,cap}} = \frac{20}{B_{\mathrm{PI}}(1)} \simeq 0.8545656833, \quad \Delta_{0,\mathrm{PI,cap}} \simeq 341.8262733 \text{ ppm}. \tag{123}$$

Below this threshold PI has strictly positive reserve; at it an occupancy reaches zero, and above it the unprotected trajectory crosses zero. Fuzzy is certified safe at $s = 1$, or 400 ppm. This proves that the Fuzzy controller has a safe work point beyond the PI directional capacity threshold at the same target, capacity and calibrated gains. Fuzzy is nonlinear in s; the proof does not assert a linear excursion law, monotone safety, or certification of every $s \leq 1$. A quantified interval around a point requires a parameter-enclosure or other uniform argument.

C.11.2 General sufficient parameter-domain interpretation. The centered initial energy is $H^2 c_P Q$, where $c_P = [1,1]\tilde{P}[1,1]^\top$ and

$$Q = \delta\omega^\top L \delta\omega = \sum_{\{i,j\} \in \mathcal{E}_u} (\omega^u_i - \omega^u_j)^2.$$

The graph counts each undirected connection once. Set $S_x = [\tilde{P}^{-1}]_{11}$ and $m_e = \min(\beta^*_e, \beta_{\max} - \beta^*_e)$. Choosing radius $R = H^2 c_P Q$, the buffer and input-domain tests are

$$H^2 c_P Q S_x R^{\mathrm{eff}}_e < m^2_e, \qquad H^2 c_P Q \Psi_i \leq r^2.$$

For a known maximum initial spread Δ_0, $Q \leq |\mathcal{E}_u|\Delta_0^2$ gives conservative direction-independent versions. These still require a valid common endpoint certificate, positive physical frequency and correction bounds; the disagreement energy alone cannot bound a common shift of all physical frequencies. They are sufficient inner-domain tests and their failure does not imply capacity failure.

Changing $\beta_{\max}$ at fixed percentage gains changes γ_i and hence the frame-count nonlinear map. The target-translation result avoids this confound by fixing $\beta_{\max}$. It also leaves the reference assumptions unchanged: nonzero link/control delays, asynchronous polling and integer frames belong to the separately evaluated Callisto model. The strict interval inclusion above concerns a compatible target slice for one prescribed frequency vector and matched gains, rather than inclusion of full robust safe sets over arbitrary initial states, networks or latency disturbances.

C.11.3 Uniform extension over a fixed parameter interval. To obtain a quantified frequency interval, fix $\mathcal{S} = [999/1000, 1001/1000]$ and cover it with 20 closed cells $\mathcal{S}_\ell = [999/1000 + \ell/10000, 999/1000 + (\ell+1)/10000]$, $\ell = 0, \dots, 19$. The intervals share endpoints and their union is exactly $\mathcal{S}$. For each cell, enclose the genuine bootstrap $x_1 = v_1 = Hs(7, -5, -2)^\top/30000$ using the same outward 10^{-60} arithmetic as the original prefix proof. Propagate all coordinates through $N = 200$ using the exact central map. No cell is discarded. Let $\overline{B}_\ell$ enclose the whole finite prefix and $\overline{V}_{\ell,N}$ its final energy. A common all-time bound is

$$B_F(s) \le \max_{\ell=0}^{19} \left\{ \overline{B}_\ell, \sqrt{\tfrac{2}{3}[\bar{P}^{-1}]_{11}\overline{V}_{\ell,N}} \right\} < 17.342 \quad (s \in \mathcal{S}). \tag{124}$$

The stored exact enclosure is approximately 17.34119237294 frames; 17.342 is an outward, simplified bound. The endpoint matrix certificate is unchanged. For the invariant-domain test use $R_{\max} = (1001/1000)^2 R_0$, and check $R_{\max}\Psi_i < 1$. The original common/deviation-frequency and correction bounds scale by at most 1001/1000 because the mean uncontrolled offset is zero and the bounds are homogeneous in its amplitude. Their exact checks preserve strictly positive physical frequency and an inactive limiter. Thus the regional formula and infinite tail apply uniformly, without assuming that a finite set of simulated parameters represents all of $\mathcal{S}$.

PI's linear scaling gives $B_{\mathrm{PI}}(s) \ge (999/1000)B_{\mathrm{PI}}(1) > 23.3802976$ for every $s \in \mathcal{S}$. Hence the rectangle

$$\mathcal{S} \times (17.342, 2048 - 17.342)$$

is a sufficient positive-headroom Fuzzy region in the (s, z) slice. At $z = 20$ every such Fuzzy trajectory has occupancy in $(2.658, 37.342)$; each PI counterpart reaches the lower boundary. This is a uniform reference-model safety separation over a fixed frequency direction and compatible target family. The method uses full interval coverage, not a monotonicity assumption on Fuzzy peak or interpolation between simulation samples. An initial unsplit enclosure was inconclusive because its tail bound exceeded 20; it is retained alongside the successful full-domain subdivision.

D Safety-Domain Validation Protocol and Complete Cases

Correspondence: Section 5. This appendix records the qualitative checks of the configuration conditions and certified comparison. The original reference/actual protocol and the uniform-interval annex were frozen separately before their respective runs. Gain selection was not part of either batch. The examples were chosen using known theory and earlier traces to test mechanisms, not to estimate an independent population advantage.

D.1 Reference configurations and verification

All cases use $\beta_{\max} = 2048$, the bidirectional triangle, $H = h = 30000000/29999$, $k_p = 2 \times 10^{-6}$, $k_i = 5 \times 10^{-11}$ and $(K_e, K_{\Delta e}, K_u) = (0.1, 29999/7500, -768/18749375)$. Memory is zero, phases are equal and $\omega_{\mathrm{ref}} = 29999/30000$. The prescribed uncontrolled offsets are $s(7, -5, -2)^\top/30000$. Link and execution delays are zero only in the reference. The continuous plant is exactly integrated between held corrections; there is no Euler approximation or replacement by a discrete firing-count plant. The reference executes the original 25-rule surface and the implemented integrate-first PI.

Twelve direct runs at $s = 1$ use $z = 16, 20, 24, 2024, 2028, 2032$. Initial occupancy, target and compatible marking all equal z. There are two cases in which both unprotected references cross, two in which both are safe, and two with strict Fuzzy safety but PI crossing, accounting for the upper/lower mirror. The point comparison gives Fuzzy's all-time interval $(z - 17.315, z + 17.315)$, whereas PI attains the exact enclosed excursion of about 23.403701 frames. The shadow virtual continuation after a first crossing is retained for diagnostics and is not a physically admissible buffer trace.

The mismatch-source grid is $s = 0.25, 0.5, 0.75, 0.85, 0.86, 1, 1.125, 1.25, 1.5, 2$. Six sources at $s = 0.5, 1, 2$ are reused only after configuration, source and raw hashes are verified; the other 14 are new. Their finite-horizon extrema determine sampled occupancy intervals by the proved translation identity. No extra plant run is charged for this derived map. Successful columns do not imply that the unsampled region is certified. The joint Lyapunov certificate is evaluated separately, including input-domain, limiter and positive-frequency conditions. The earlier midpoint grid certifies PI through $s = 16$ and Fuzzy through $s = 2$ under their respective coarse tests; this is not a true safe-set ordering.

Four further runs test $(6, -6, 0)/30000$ and $(8, -4, -4)/30000$ at $z = 1024$. Alongside the prescribed direction, their pairwise initial spreads are all 400 ppm and their means coincide. However, $Q = \delta\omega^\top L \delta\omega$ takes values $(216, 234, 288)/30000^2$. The exact energy identity is checked; PI's peaks coincide here because of the triangle's repeated disagreement eigenvalue, while Fuzzy's peaks differ. Thus the energy condition is sufficient and distribution-aware, without predicting a strict ordering of measured peaks.

The separately frozen annex adds four runs at $s = 0.999, 1.001$, $z = 20$. It independently recomputes every outward-arithmetic cell of the uniform proof. Fuzzy's endpoint minima are approximately 2.703366 and 2.668520; PI's are −3.380298 and −3.427105. The endpoint runs check implementation consistency. The 20-cell coverage and infinite-tail argument, not endpoint interpolation, establish the all-time statement over the entire interval.

All reference runs use bootstrap plus 2000 subsequent intervals. Maximum translation residual is 6.83×10^{-13} frame and maximum phase residual 2.78×10^{-12}. The original comparison tolerances are unchanged. Interpolation and rule replay are checked separately from the exact proof certificate. The record contains 34 new reference trajectories, six cached sources and separate proof/postprocessing evidence.

D.2 All delayed Callisto cases

The actual runs restore physical link latency 20, local polling 1000, execution delay 10 and integer events over $T = 2001H$. They fix

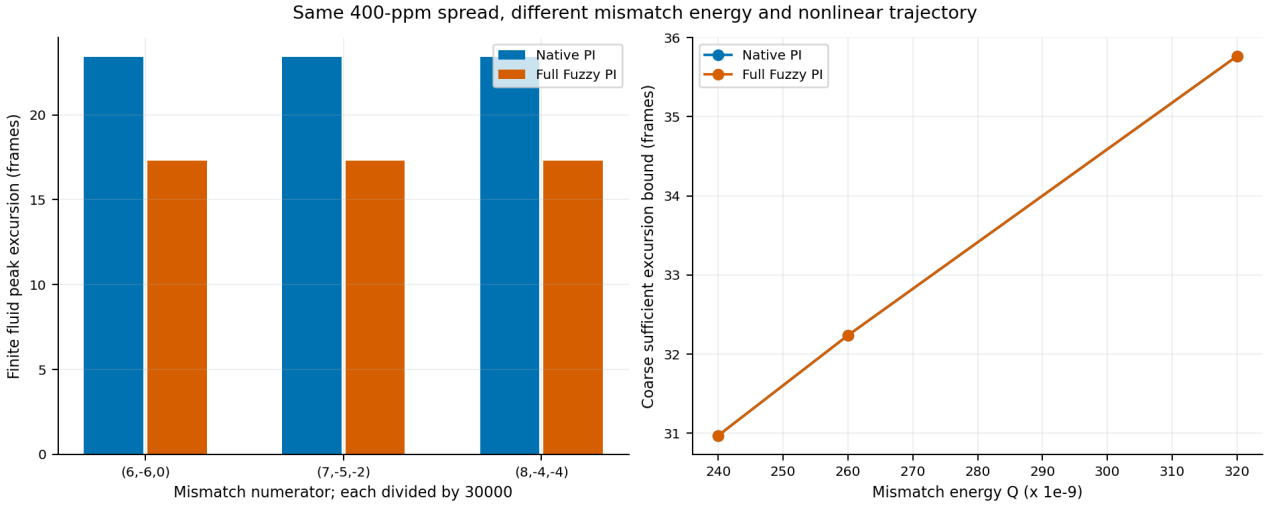


Figure 4: Three equal-spread initializations with different mismatch energy. The energy identity is exact; a larger certificate radius does not imply a larger exact trajectory peak.

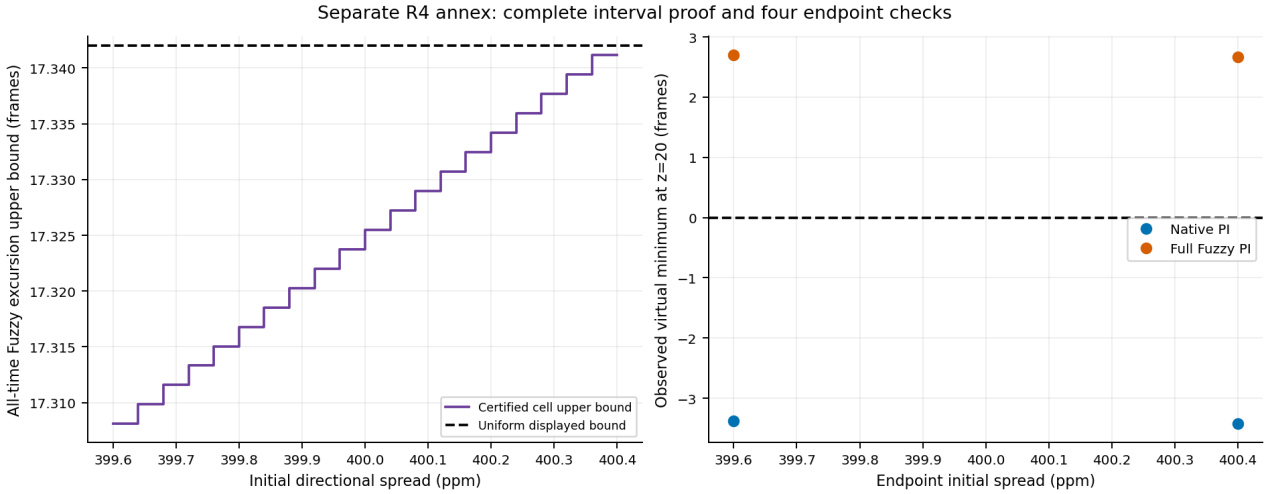


Figure 5: Complete parameter-cell proof and direct endpoint checks. All 20 cells cover 399.6–400.4 ppm at the fixed mismatch direction. The maximum Fuzzy all-time bound is below 17.342; PI's minimum attained peak exceeds 23.3802976 throughout the interval.

$s = 4$, initial phase 0.1, all capacities and gains above, and use the original negative-time source history. For each z, PI's target is z, Fuzzy's percentage setpoint is $100z/2048$, and the marking is initial occupancy plus initial in-flight frames. Each paired history and marking is identical. Guards are enabled with reserve zero; observer, supervisory switching and anti-windup are disabled. Requested/resolved settings are checked before results are used.

At $z = 83$ PI's first input-empty guard occurs at $t \simeq 71844.671848$ ticks; five episodes sum to about 61.0114 blocked node-ticks. Fuzzy remains unstalled. The physical clock continues to advance when data phase pauses, so physical-phase and data-phase progress must be recorded separately.

Upper-target cases do not reproduce the receiving-buffer mirror of the zero-delay reference. In this runtime, used outgoing credit is the UGN offset plus source sends minus causally delayed reverse pops. Its initial value is $z + 40$ although initial receiving occupancy is z: the accounting includes forward in-flight data and delayed reverse consumption knowledge. Full credit replay verifies its capacity constraint. All upper-target cases guard while the maximum physical receiving occupancy remains 2008. The first credit guards at $z = 1944$ occur at approximately 47886.861480 ticks (PI) and 57745.333282 ticks (Fuzzy), matching the corresponding $z = 64$ lower-target cases. This explains the discrepancy without changing gains. The observed 40-frame difference is specific to the declared

Table 7: All actual qualitative validation cases. Occupancy extrema include every integer event. Blocked time sums disabled-data-phase duration across nodes, rather than measuring elapsed network downtime.

z	Controller	Buffer range / frames	Off episodes	Blocked node-ticks	First guard cause
104	PI	10–198	0	0	none
104	Fuzzy	32–176	0	0	none
83	PI	0–166	5	61.0114	empty input
83	Fuzzy	11–155	0	0	none
64	PI	0–128	26	131.0456	empty input
64	Fuzzy	0–128	6	47.0093	empty input
1944	PI	1880–2008	26	131.0456	delayed outgoing credit
1944	Fuzzy	1880–2008	6	47.0093	delayed outgoing credit
1965	PI	1922–2008	51	222.1022	delayed outgoing credit
1965	Fuzzy	1922–2008	3	84.0205	delayed outgoing credit
1984	PI	1960–2008	5837	587.8914	delayed outgoing credit
1984	Fuzzy	1960–2008	1198	188.3495	delayed outgoing credit

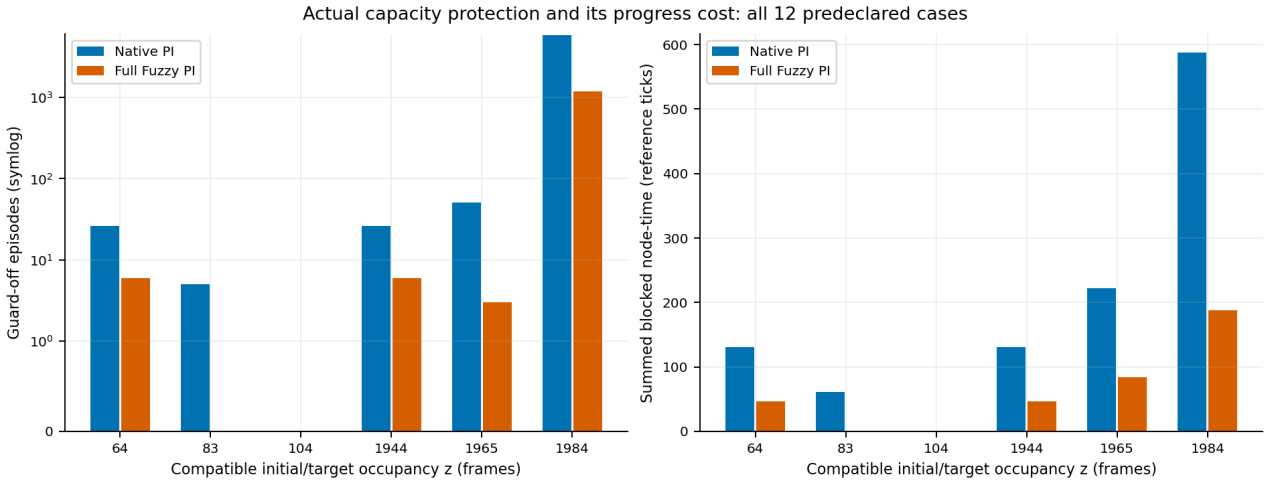


Figure 6: Guard-off episodes and summed blocked node-time for every predeclared actual case. Three cases remain unstalled. Nine retain capacity with guard intervention; counts alone are not throughput.

constant latency, history and protocol; it is not a general delay margin or an all-time theorem for Callisto.

Every actual run has positive physical frequency (minimum 0.9993) and no saturated correction. Complete integer arrival/pop replay has zero occupancy mismatches; independent PI/25-rule command reconstruction has maximum residual 1.09×10^{-19}. Pre-first-guard measurements, command times and corrections agree with the high-reserve source trajectory. Subsequent evolution is from independent actual runs, not shifted old curves. All 12 runs remain within inclusive physical capacity; the nine guarded cases do not establish controller-only strict interior safety. At $z = 1965$, Fuzzy has fewer guard episodes but lower final data progress than PI, so neither episode count nor blocked duration alone demonstrates a throughput improvement.

D.3 Why delayed credit changes the upper constraint

For this constant-latency runtime, define the data-phase count $N_i(t) = \lfloor \psi_i(t) \rfloor - \lfloor \psi_i(0) \rfloor$, including its declared negative-time extension. Let τ^{f}_{ji} be forward delivery latency and τ^{a}_{ij} be the reverse consumption-feedback latency. The compatible relative marking λ_{ji} accounts for initial receiving occupancy and initial in-flight frames. Ordered constant-latency delivery and causal reverse-pop feedback give

$$\begin{aligned} \beta_{ji}(t) &= \lambda_{ji} + N_j(t - \tau^{\mathrm{f}}_{ji}) - N_i(t), \\ \chi_{ji}(t) &= \lambda_{ji} + N_j(t) - N_i(t - \tau^{\mathrm{a}}_{ij}). \end{aligned} \tag{125}$$

Subtracting these count balances yields

$$\chi_{ji} - \beta_{ji} = \underbrace{N_j(t) - N_j(t - \tau^{\mathrm{f}}_{ji})}_{n^{\mathrm{flight}}_{ji}(t)} + \underbrace{N_i(t) - N_i(t - \tau^{\mathrm{a}}_{ij})}_{n^{\mathrm{unack}}_{ji}(t)}. \tag{126}$$

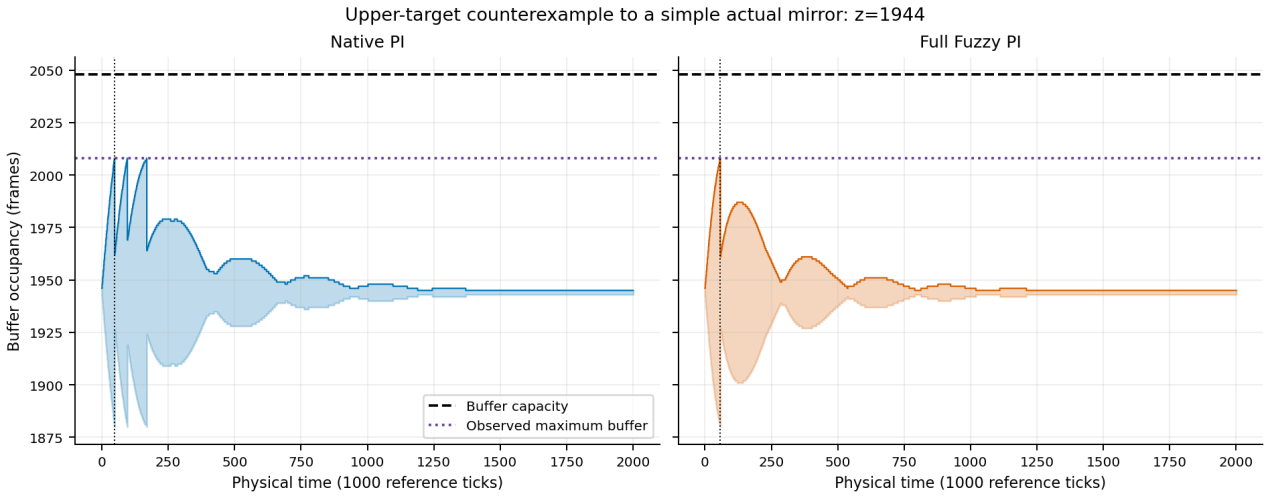


Figure 7: Actual upper-target credit constraint at $z = 1944$. Credit readiness can stop sending before receiving occupancy reaches capacity. Forward in-flight frames and delayed reverse-pop knowledge prevent an occupancy-only upper/lower mirror argument.

The first count represents forward in-flight frames; the second represents consumed frames whose credit has not reached the sender. Both are nonnegative for monotone data phase. Hence $\chi_{ji} \geq \beta_{ji}$, and $\beta_{ji} < \beta_{\max}$ does not imply that one more frame can be sent. This algebraic runtime identity explains the observed guards; it is not an additional all-time stability certificate.

At the declared initialization, both latencies are 20 and the negative histories yield 20 frames in each count, so $\beta_{ji}(0) = z$ while $\chi_{ji}(0) = z + 40$. Afterwards these counts depend on phase histories and guard pauses; they cannot generally be replaced by a fixed 40-frame allowance. The identity applies to the constant-latency protocol tested here. Variable-latency ordered transport requires its actual arrival and acknowledgment maps, rather than this fixed time-shift expression.

D.4 Claim–evidence separation

Exact matrix/interval calculations certify the reference inner domain, strict target-interval benefit and uniform narrow directional interval. Reference simulation checks the original rules and model identities. Delayed actual runs check whether the margin mechanism persists, and expose an additional credit condition. They do not certify arbitrary topology, delay, initial error at a fixed target, globally optimal PI comparison, or application-payload logical synchrony. Earlier random, burst and independently tuned comparisons remain separate supplementary evidence in Appendix E.

E Experimental Protocol and Supplementary Evidence

Correspondence: Section 5. This appendix preserves the extended model, intermediate steps and supporting interpretation behind the corresponding main-text section.

The evaluation connects the controller analysis to three distinct questions: whether the continuous-clock implementation respects the PI calibration; whether the nonlinear reference trajectories agree with the mathematical analysis; and whether the proposed controller offers a measurable benefit against independently tuned PI under shared delivery disturbances. We separate matched-gain mechanism checks from equal-budget controller selection, and reference-model certificates from Callisto simulation measurements.

E.1 Experimental design and continuous-clock execution

All Callisto simulations use the continuous-phase Callisto realization of Section B. Clock phases evolve continuously, integer frame crossings determine buffer events, and measurements and correction applications are scheduled by each node's local phase. There is no common physical polling grid. Table 9 gives the nominal plant; the startup stress study changes the uncontrolled frequencies and reconstructs the associated negative-time histories explicitly.

Evidence groups and data separation. The implementation check, M0, contains six matched pairs, or 12 actual runs. An earlier fixed-preset burst campaign, M2, supplies a separate mechanism diagnostic. The earlier fixed-preset random campaign, M3, is development evidence and supplies no seeds to the new independent tests. Its fixed-preset comparison did not support lower Fuzzy RMS against moderate PI; the new study asks the different question of independently selected controllers under an equal evaluation budget. The new training study charges 192 slots, including four verified cached M2 evaluations and 188 new Callisto simulation runs. Selection is frozen before 186 new held-out runs. A separately registered slower-disturbance study adds 107 Callisto simulation runs, including its engineering pilots, and the startup study adds six. Its 32 synchronous fluid-reference trajectories are reference calculations,

Table 8: Theoretical predictions and their numerical checks. Finite simulation checks consistency; the stated infinite-horizon conclusions come from the corresponding mathematical arguments.

Prediction	Verification	Interpretation
Compatible-target translation	12 direct nonzero-target runs; maximum occupancy residual 6.83×10^{-13}	Fixed $\beta_{\max}$ and gains; initial occupancy, target and marking change together.
Directional PI capacity threshold	At $z = 20$, $s = 0.85$ gives excursion 19.893146; $s = 0.86$ gives 20.127183	These bracket the proved 341.8263-ppm threshold along one direction.
Strict reference safety separation	$z = 16$: both cross; $z = 20$: only PI crosses; $z = 24$: both safe; upper mirrors agree	Full-time guarantees follow from certified prefix/tail bounds, not record length.
Uniform parameter interval	All 20 outward-arithmetic cells cover 399.6–400.4 ppm; four direct endpoints agree	The full cells and tail prove the interval; endpoint interpolation does not.
Distribution-dependent mismatch energy	Three equal-spread directions give $Q = (216, 234, 288)/30000^2$ and satisfy $V_1 = H^2 c_P Q$	Equal spread need not give equal energy; energy is not an exact-peak ordering.
Finite delayed execution benefit	At $z = 83$, Fuzzy retains minimum 11 and no guard; PI reaches 0 and invokes five episodes	Same configuration and matched gains; PI protection avoids invalid consumption.
Additional delayed readiness constraint	Every upper-target case invokes credit guards while maximum receiving occupancy is 2008	Equation (126) identifies the missing transit/feedback state.

Table 9: Nominal Callisto simulation configuration. Time is in physical reference ticks except where local ticks are explicitly stated. Controller parameters selected for independent tests are listed separately in Table 11.

Parameter	Value
Topology	Three-node bidirectional triangle; six receiving buffers
Uncontrolled frequencies ω^u	$(1.0002, 0.9998, 0.9999)^\top$; initial spread 400 ppm
Initial phase; base frequency	0.1 at every node; $\omega_{\text{base}} = 29999/30000$
Capacity; initial occupancy; target	$\beta_{\max} = 2048$; $\beta_e(0) = \beta_e^* = 1024$ frames
Initial in-flight frames; marking	20 per channel; 1044 in the recorded convention
Nominal link latency	20
Polling; control-execution delay	$p = 1000$; $d_{\text{ctrl}} = 10$ local ticks
Execution constraints	$\lvert c_i \rvert \le 0.01$; guards enabled; reserve zero; no anti-windup
Control law	Implemented integrate-first PI baseline or the complete incremental 25-rule Fuzzy law
Other mechanisms	Observer and supervisory mode switching disabled
Disturbance onset	$t_{\text{on}} = 2 \times 10^6$

not additional Callisto simulation runs. Old data, infeasible candidates and repair records are retained in the evidence archive.

Shared delivery inputs. All controllers receive identical externally generated desired-latency profiles within each comparison, verified by content hashes. Delay is 20 before onset, including the negative-time prehistory. Ordered arrivals use the causal running-maximum envelope of Section B; a delay decrease may release an ordered batch. The model does not impose a finite serialization rate. For bounded random latency, the desired delay is $20 + z_m$, where

$$\begin{aligned} z_0 &= 5\varepsilon_0, \\ z_m &= r_z z_{m-1} + (1 - r_z)5\varepsilon_m, \quad r_z = e^{-1}, \quad \varepsilon_m \sim \mathcal{U}[-1, 1]. \end{aligned} \tag{127}$$

Thus desired delay remains in $[15, 25]$. The index m counts physical profile updates on a grid of spacing T_c, independently of controller polling. Here T_c denotes a profile-update interval, not an exactly measured correlation time. The original two regimes use $T_c = 1000$ and 10000; the later confirmation study uses 100000.

E.2 Metrics, admissibility and comparison windows

For n nodes, let $\bar{\omega}(t) = n^{-1} \sum_i \omega_i(t)$ and define the instantaneous root-mean-square (RMS) deviation of node frequencies from their network mean as

$$\begin{aligned} \sigma_\omega(t) &= \sqrt{n^{-1} \sum_i [\omega_i(t) - \bar{\omega}(t)]^2}, \\ \sigma_{\text{ppm}}(t) &= 10^6 \sigma_\omega(t)/\omega_{\text{unit}}, \qquad \omega_{\text{unit}} = 1. \end{aligned} \tag{128}$$

Parts per million (ppm) use the fixed normalized frequency unit, not ω_{base}. For a physical-time window $W = [a, b)$ of duration $T_W = b - a$, report

$$\begin{aligned} R_W &= \sqrt{T_W^{-1} \int_W \sigma_{\text{ppm}}(t)^2 \, dt}, \\ I_W &= \int_W \sigma_{\text{ppm}}(t) \, dt, \qquad P_W = \sup_{t \in W} \sigma_{\text{ppm}}(t). \end{aligned} \tag{129}$$

RMS inter-node frequency deviation refers here to deviation about the instantaneous network mean, rather than tracking a fixed external reference frequency. The integral absolute error (IAE) I_W integrates RMS amplitude, rather than the mean absolute node deviation. Frequency moments use complete held-frequency intervals weighted by their physical durations. The actual pairwise spread $\Delta_\omega = \max_i \omega_i - \min_i \omega_i$ is recorded separately. Although $\Delta_\omega \le \sqrt{2n}\,\sigma_\omega$ pointwise, window RMS alone does not establish a maximum transient-spread bound.

Buffer excursion, minimum two-sided headroom and mean command total variation (TV) are

$$B_W^{\mathrm{obs}} = \max_e \sup_{t\in[a,b]} |\beta_e(t) - \beta_e^*|,$$

$$H_W^{\mathrm{obs}} = \min_e \inf_{t\in[a,b]} \min\{\beta_e(t), \beta_{\max} - \beta_e(t)\}, \tag{130}$$

$$\mathrm{TV}_W = n^{-1} \sum_i \sum_{k:\, a\le t_{i,k}^a \le b} |c_{i,k} - c_{i,k-1}|. \tag{131}$$

Buffer statistics reconstruct all integer crossings and ordered arrivals, grouping simultaneous events and including the endpoint state. Sparse plot samples do not determine extrema. TV includes both endpoints under the frozen convention, so adjacent-window activities cannot be added without treating the shared endpoint. TV measures command activity, not hardware energy. Requested and applied corrections, saturation, gate events, positive frequency and data-phase progress are checked separately.

Before each post-centering disturbance, three consecutive 10^5-tick windows ending at onset must satisfy RMS inter-node frequency deviation at most 2 ppm, each buffer's absolute mean bias at most two frames and standard deviation at most one frame, common-frequency temporal standard deviation at most 1 ppm and three-window mean range at most 1 ppm, with positive progress and no intervention. These are finite engineering centering criteria, not exact equilibrium. A training candidate failing the same criteria in any required case remains infeasible and consumes its assigned budget.

The original-regime tests use $W = [2\times10^6, 4\times10^6)$ and tail $[3.6\times10^6, 4\times10^6)$. Each controller's own clean trajectory is its same-parameter, nominal-delay counterfactual over the same physical times. Common-frequency shift is reported relative to this counterfactual, separately from relative synchronization. Strict recovery requires the centering criteria for three consecutive windows and a window-mean common frequency within 2 ppm of own clean. A return not observed by the specified endpoint is right-censored, not assigned a recovery time of zero or interpreted as permanent nonrecovery.

E.3 Implementation and nonlinear-reference validation

M0 checks Proposition B.1 by replacing the Fuzzy surface with its calibrated linear counterpart F_0, without changing the continuous plant. Six pairs cover polling periods 100 and 1000, initial frequency spreads 35 and 400 ppm, and capacity/target settings (2048, 1024), (512, 128) and (512, 384) over 2×10^5 ticks. Across 19,800 paired updates, the maximum correction difference is 5.29×10^{-18}; physical/data phases, measurement vectors and saved buffer counts agree, without saturation or guard intervention. This checks initialization, integration order and gain mapping. It does not replace the complete nonlinear law in the performance experiments.

The original 25-rule law is also evaluated directly on the synchronous, zero-latency reference of Section C, with 2000 post-bootstrap updates. At the prescribed moderate-gain initialization, the fluid buffer peak is approximately 17.314056818 frames, maximum $|X| + |Y|$ is 0.090130790, maximum absolute correction is 3.21398377×10^{-4}, and no contraction-envelope violation is observed. Physical frequencies stay in $[0.9998, 1.0002]$ including startup. These finite numerical checks agree with the analysis; the all-time comparison, Fuzzy below 17.315 versus PI approximately 23.403701 frames, follows from the interval-prefix and contracting-tail proof in Theorem C.4.

Rule verification. We reconstruct requested corrections from the saved inputs in all 299 new validation, slow-delivery and Callisto startup records. The maximum residual is approximately 5.42×10^{-20}. This uses the complete rule map and stored accumulator, not a sector surrogate. The earlier M2 diagnostic additionally stays within Ω_5, with observed $\max(|X| + |Y|) = 2.001888$, supporting application of Proposition B.2 to those measured inputs. This diagnostic is retained separately from the independently selected controllers: Fuzzy_B reaches approximately 39.159 in the large held-out burst. The full 25-rule law remains defined there, whereas the central-region identity and the Ω_1 reference witness do not cover that whole trace. Same-input memory accounting is an algebraic diagnostic, not an additional closed-loop F_0 experiment.

E.4 Reference safety strata and Callisto startup transients

To vary initial mismatch without changing occupancy normalization, capacity and targets remain fixed while

$$\omega^u(s) = \frac{29999}{30000}\mathbf{1} + s\frac{(7,-5,-2)^\top}{30000}, \quad s \in \{0.5, 1, 2, 4, 8, 16, 32, 64\}. \tag{132}$$

Here s is an initial-mismatch scale, with uncontrolled spread $400s$ ppm. The reference uses the matched moderate coordinates $(k_p, k_i) = (2\times10^{-6}, 5\times10^{-11})$, synchronous zero-latency fluid execution and $H = h = 30000000/29999$. Its 32 trajectories comprise two controllers, eight scales and both unprotected virtual and 0.01-clipped variants. Each includes the bootstrap and 2000 subsequent zero-order-hold intervals. The initial certificate energy scales as $V_1(s) = s^2 V_1(1)$; input-domain, buffer, positive-frequency and correction conditions are checked jointly.

The largest grid point satisfying the common sufficient conditions is $s = 2$. At $s = 4$, both reference trajectories remain normal over the record although the certificate no longer guarantees them. At $s = 32$, unprotected reference requests first require actuator clipping; at $s = 64$, both virtual reference buffer trajectories violate capacity. Figure 8 distinguishes measured virtual peaks from sufficient bounds. Failure of a sufficient condition is inconclusive, rather than a prediction of instability.

Before running Callisto cases, the selection rule freezes three common scales: 2, 4 and 32, representing certified, observed-safe uncertified, and correction-constrained reference strata. Callisto restores link latency 20 and local polling/control delay 1000/10. Its

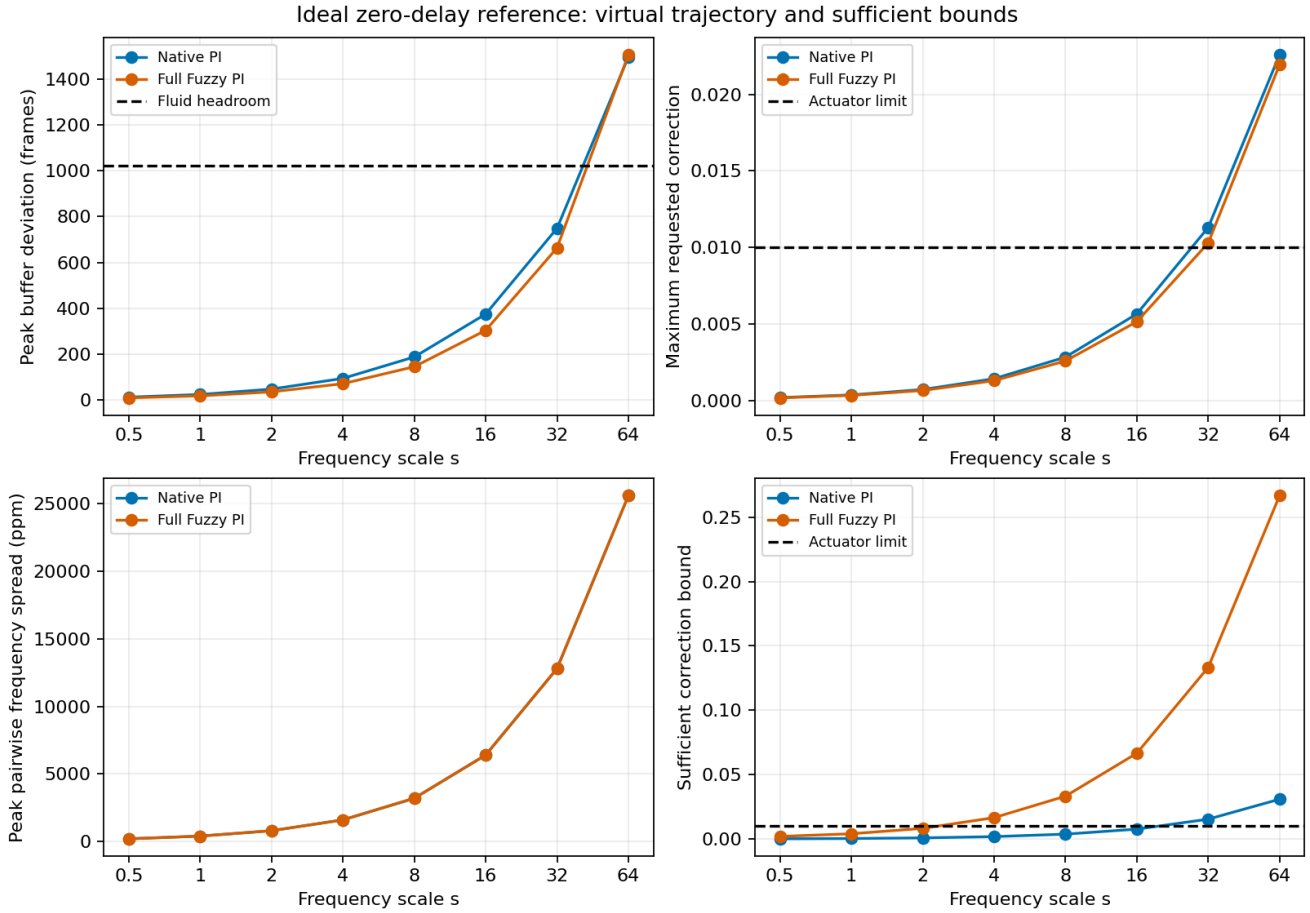


Figure 8: Synchronous zero-latency fluid-reference sweep at matched moderate coordinates. Virtual trajectory peaks and analytical sufficient bounds are distinct quantities. The dashed levels show fluid buffer headroom and the correction limit; clipping and unprotected virtual trajectories are archived separately. These are reference calculations, not actual Callisto safety certificates.

Table 10: Callisto startup stress at matched moderate coordinates. Buffer excursions include startup and all reconstructed events. Sat. counts saturated control updates; every case has zero gate events and replay mismatches.

		Buffer excursion / frames		Sat.
s	Spread / ppm	PI	Fuzzy	PI / Fuzzy
2	800	47	37	0 / 0
4	1600	94	72	0 / 0
32	12800	750	664	176 / 85

locked startup horizon is $2001H \simeq 2.001067 \times 10^6$ physical ticks, rather than the 4×10^6 used for the disturbance tests. This shorter horizon is an execution-plan deviation from the earlier completion instruction and limits the empirical safety observation accordingly. The six cases are startup stress tests, not post-centering disturbance tests.

Fuzzy has a smaller maximum actual buffer excursion at all three selected scales (Table 10). At $s = 4$, the reduction is from 94 to 72 frames, approximately 23.4%, with no saturation or gate intervention. Figure 9 shows the associated frequency and occupancy transients. All six records retain positive frequency, data progress and capacity compliance. At $s = 32$, Fuzzy also requests fewer saturated updates, but 12800 ppm is a stress setting, not a typical hardware tolerance. Its histories are initialized consistently: the faster source has 21 in-flight frames and marking 1045, whereas the other sources retain 20 and 1044. Both controllers receive identical histories at each scale.

These observations show a smaller transient buffer demand at matched gains, but do not transfer the reference theorem to delayed, asynchronous, quantized execution. The recorded pairwise spread remains a separate quantity. No hardware-based numerical Δ_{safe} is assumed; applying the reference spread corollary would require an independently justified threshold exceeding $D(R)$.

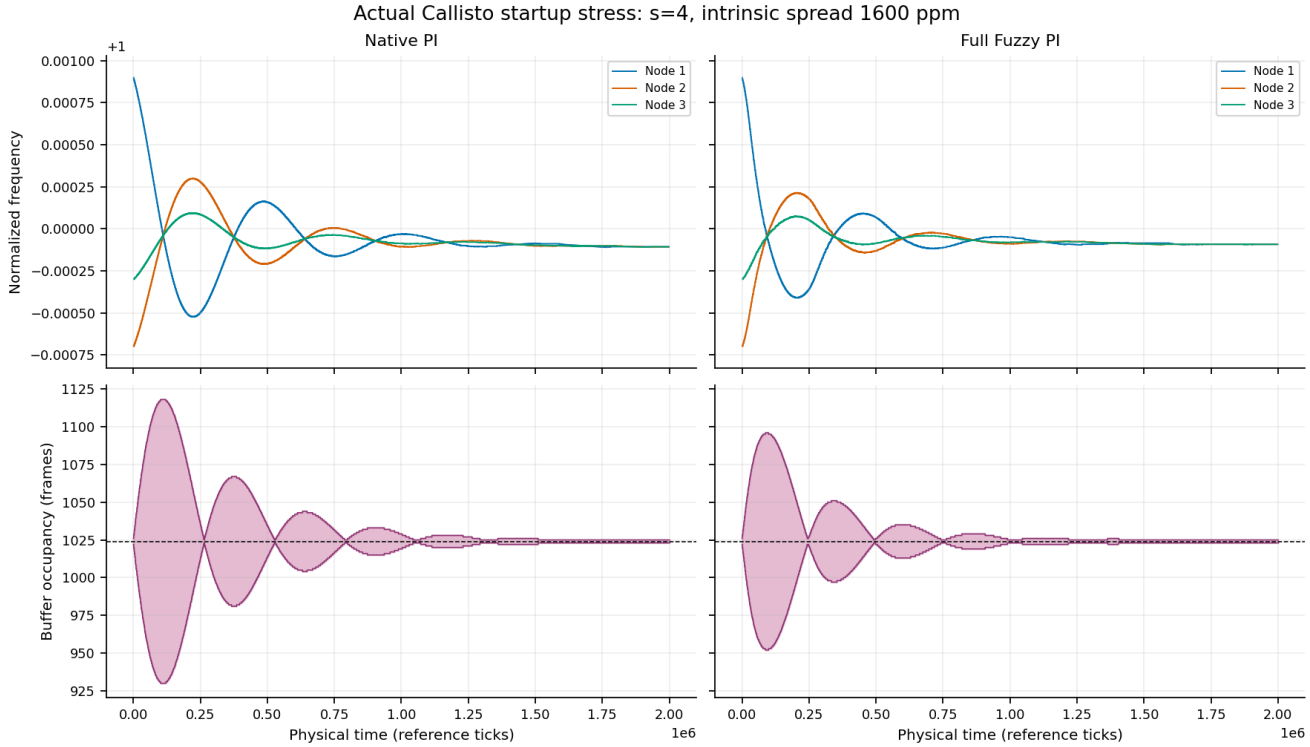


Figure 9: Continuous-phase Callisto startup at $s = 4$ (1600 ppm), with matched moderate coordinates. Upper panels show positive absolute normalized frequencies using a +1 axis offset. Lower panels show the minimum–maximum occupancy envelope across all receiving buffers, from the complete crossing stream. Fuzzy reduces the maximum target deviation from 94 to 72 frames over the stated startup horizon.

E.5 Equal-budget selection and independent random-latency tests

Fixed search budget and selection. Each family uses the same 24-point grid,

$$k_p \in \{1.5, 2, 2.5, 3, 4, 6\} \times 10^{-6},$$
$$k_i \in \{1, 2, 5, 10\} \times 10^{-11}. \tag{133}$$

Fuzzy fixes $K_e = 0.1$ and obtains $K_{\Delta e}, K_u$ from the calibration in Section B; it receives no extra shape-search dimension. Four training cases per candidate comprise clean operation, a single-edge 20-tick delay increment lasting 10^5 ticks, and two random regimes using training seed 31001. Both families therefore receive 96 evaluation slots. Four hash-verified cached evaluations remain charged within the 192-slot total. Nineteen PI and 22 Fuzzy candidates satisfy the same finite-run admissibility requirements; all seven infeasible candidates are retained without replacement.

Five training objectives are minimized separately: worst random-case RMS, burst IAE, worst disturbance-window buffer excursion, peak absolute common-frequency excursion from own clean, and worst mean command TV. Each family's feasible Pareto set supplies a random-priority representative, labeled R, and a burst-priority representative, labeled B. The frozen lexicographic rule prioritizes random RMS for R and burst IAE for B, with the remaining objectives and candidate identifier as declared tie breakers. One PI serves both priorities; Fuzzy selects distinct representatives (Table 11). Deduplication gives three candidates, without backfilling.

Held-out paired tests. Only after freezing selection are test profiles generated from seeds 41001–41030. All three candidates execute both original random regimes and both held-out burst combinations, giving $30 \times 2 \times 3 + 2 \times 3 = 186$ new Callisto simulation runs. All 186 complete and satisfy the declared normal-admissibility checks, with no engineering QA errors or complete-event replay mismatches. No seed pair is excluded. Archived M3 seeds are not reused as held-out tests.

The primary random comparison is Fuzzy_R versus PI_R. Let ℓ index independent seeds. Report the paired mean difference and mean of per-seed ratios,

$$\widehat{d} = \frac{1}{30} \sum_{\ell=1}^{30} (R_{W,\ell}^{\text{Fuzzy}_\text{R}} - R_{W,\ell}^{\text{PI}_\text{R}}),$$
$$\widehat{r} = \frac{1}{30} \sum_{\ell=1}^{30} R_{W,\ell}^{\text{Fuzzy}_\text{R}} / R_{W,\ell}^{\text{PI}_\text{R}}. \tag{134}$$

The ratio is not the ratio of group means. Confidence intervals (CIs) use 20000 paired-seed percentile-bootstrap resamples, with fixed resampling seeds 58001/58002. Pointwise 95% intervals accompany each endpoint; a joint statement about the two original regimes also

Table 11: Frozen representatives. The k_p, k_i columns are PI baseline gains for PI and calibrated PI coordinates for Fuzzy. R and B denote random- and burst-priority selection; the candidates are not retuned on tests.

Representative	Candidate	k_p	k_i	K_e	$K_{\Delta e}$	K_u
$\mathrm{PI}_R = \mathrm{PI}_B$	PI_g05	2×10^{-6}	10^{-11}	—	—	—
Fuzzy_R	Fuzzy_g02	1.5×10^{-6}	2×10^{-11}	0.1	7.499750	-1.638455×10^{-5}
Fuzzy_B	Fuzzy_g17	4×10^{-6}	10^{-11}	0.1	39.998667	-8.192273×10^{-6}

uses nominal Bonferroni-adjusted 97.5% intervals per regime. Time points, edges and bootstrap draws are not independent replications.

Table 12 and Figure 10 show reductions in RMS inter-node frequency deviation at both tested profile grids. The mean paired relative improvements are 16.78% and 7.85%. Both nominal family-adjusted intervals lie below zero.

For the first regime, the interval is $[-0.81274, -0.73483]$ ppm. For the second, it is $[-0.44451, -0.28407]$ ppm. This supports an advantage of the selected Fuzzy scheme under the declared finite-grid budget, plant and two input regimes. Because the selected coordinates differ between families, it does not isolate a same-gain inference-only effect or establish globally optimal PI/Fuzzy performance.

Activity and common-frequency cost. Mean per-node command TV decreases from 0.0102361 to 0.00782282 at $T_c = 1000$, and from 0.00112361 to 0.000906694 at $T_c = 10000$ (PI_R to Fuzzy_R). All selected candidates have at most six frames of disturbance-window target deviation. However, Fuzzy_R has larger magnitude mean tail common-frequency shifts from own clean: −61.420 versus −15.692 ppm, and −31.933 versus −15.384 ppm. Thus smaller inter-node frequency differences and command activity do not imply improved absolute-frequency regulation. No absolute-frequency reference loop is present.

E.6 Held-out bursts and slower delivery changes

Burst-priority validation. The two predetermined held-out combinations are a 10-tick desired delay increment lasting 10^4 ticks on channel $2 \to 3$, and a 40-tick increment lasting 10^5 ticks on $3 \to 1$, both beginning at 2×10^6. They are distinct from the training combination. For the larger burst, Fuzzy_B lowers post-onset IAE from 8.19241×10^6 to 3.62317×10^6 ppm·ticks, approximately 55.8%, while matching the 40-frame buffer excursion. Table 13 retains both deterministic cases: the improvement depends on the burst combination rather than holding for every tested burst. These two cases do not supply seed-level CIs.

Ramp–hold–ramp inputs. A separate registered study freezes the selected gains and applies a 20-tick delay increment on channel $1 \to 2$ using rise/fall times $T_r \in \{10^4, 10^5\}$ and hold times $T_h \in \{10^5, 10^6\}$. The four combinations all run to 5.2×10^6 ticks. The end of the falling ramp is $t_{\mathrm{off}} = 2 \times 10^6 + 2T_r + T_h$; each case has its own $[t_{\mathrm{off}}, t_{\mathrm{off}} + 2 \times 10^6)$ recovery window and same-horizon clean counterfactual. A larger constant delay plateau does not mean delivery stops throughout that plateau. The ramp instead gives feedback time to act while the delivery imbalance forms.

For $T_r = 10^5$, maximum buffer deviations are 19 frames for PI, 18 for Fuzzy_R and 17 for Fuzzy_B. The slower formation of the disturbance therefore provides another observed buffer-margin benefit. Frequency RMS favors PI across the four ramp combinations, and strict own-clean recovery is censored in all 12 ramp records. Different disturbance durations are not pooled into a raw-IAE ranking; these data support a resource/frequency tradeoff, not a faster-settling claim.

Separately registered slow-random confirmation. The slower grid $T_c = 100000$ uses new seeds 51001–51030, unchanged selected gains, the same bounded recursion, and $W = [2 \times 10^6, 5 \times 10^6)$ with tail $[4.6 \times 10^6, 5 \times 10^6)$. Each trace has 30 complete profile blocks, which are not independent statistical replicates. The study includes 105 matrix slots: three long-clean cases, 12 ramps and 90 random cases. Four matrix slots reuse qualified pilot runs; 101 new main runs plus six pilots give 107 unique actual runs. All matrix slots pass profile, nominal-prefix and event-replay checks, with no excluded confirmation seed pairs.

This confirmation bounds the applicability of the original-regime advantage: the mean paired $\mathrm{Fuzzy}_R/\mathrm{PI}_R$ RMS ratio is 1.33312 with pointwise 95% CI [1.30337,1.36219], using 20000 paired bootstrap resamples and fixed seed 59001. The mean paired difference is +1.53215 ppm [1.39578,1.66175]. Slower variation alone therefore does not extend the selected controller's RMS advantage. This confirmation delimits the two original-regime benefits without changing their registered primary endpoints or the frozen controller selection.

E.7 Connection between analysis and measured benefits

The evidence supports two benefits. First, nonlinear analysis at matched gain coordinates proves a strict headroom improvement for the prescribed fluid reference; Callisto startup tests at matched gains also show smaller measured excursions at three fixed initial mismatches. Second, equal-budget, independent selection yields lower RMS inter-node frequency deviation and mean command activity in two held-out random regimes, while the burst-priority representative reduces IAE in the larger held-out burst. These are distinct statements with different gains, assumptions and observation windows.

Their scope is explicit: common-frequency shift remains a cost, the small burst and slower delivery tests delimit performance, and finite Callisto records do not certify all-time safety or convergence. The experiments do not establish a general safe-set inclusion, a globally optimal comparator, payload-level functional logical synchrony, or hardware energy savings. Positive phase progress and buffer compliance show normal model execution; they are not a

Table 12: Held-out original-regime tests: 30/30 admissible pairs per row. Controller RMS entries are seed means in ppm. Difference is Fuzzy$_R$ minus PI$_R$; ratios are means of paired ratios. Brackets give pointwise 95% paired bootstrap intervals.

Profile grid	PI$_R$ RMS	Fuzzy$_R$ RMS	Difference / ppm [CI]	Paired ratio [CI]
$T_c = 1000$	4.61355	3.83949	−0.77406 [−0.80767,−0.73992]	0.83217 [0.82484,0.83964]
$T_c = 10000$	4.65204	4.28718	−0.36486 [−0.43523,−0.29464]	0.92150 [0.90630,0.93672]

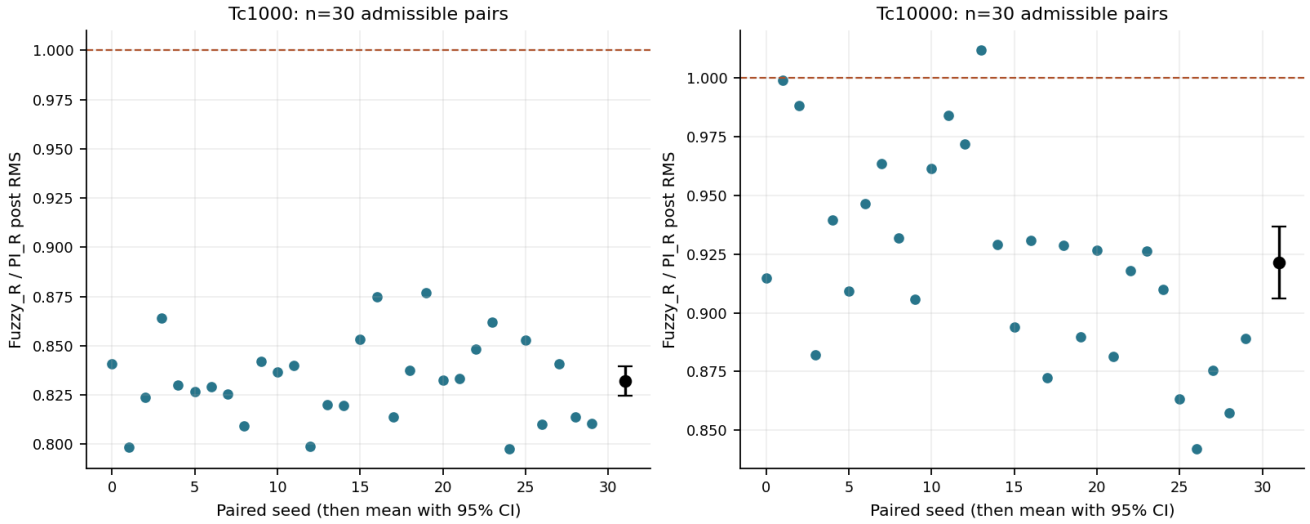


Figure 10: Independent paired Fuzzy$_R$/PI$_R$ RMS effects after equal-budget selection. Each dot is one of 30 held-out seed pairs. The final black marker is the mean paired ratio with a pointwise 95% CI; the dashed line denotes equal RMS. Individual seed outcomes and their aggregate are both retained.

Table 13: Both held-out burst-priority cases over $[2 \times 10^6, 4 \times 10^6)$. IAE is in 10^6 ppm·ticks; buffer excursion is in frames. A and D denote desired delay increment and its duration, respectively.

	IAE		Buffer excursion	
Case	PI$_B$	Fuzzy$_B$	PI$_B$	Fuzzy$_B$
$A = 10, D = 10^4$	0.265814	0.352614	11	11
$A = 40, D = 10^5$	8.192410	3.623166	40	40

functional refinement proof. Within these limits, the combined analysis and tests provide both a mechanistic reason and independent empirical evidence for considering the proposed Fuzzy PI law when transient buffer use and disturbance-dependent synchronization performance matter.

F Extended Interpretation and Scope

Correspondence: Section 6. This appendix preserves the extended model, intermediate steps and supporting interpretation behind the corresponding main-text section.

The results identify two reasons to consider the proposed Fuzzy PI law for bittide: a provable reduction in transient buffer demand for a specified reference execution, and independently tested synchronization benefits in defined delivery regimes. The connection between these results is the explicit nonlinear correction mechanism, rather than an assumption that fuzzy inference improves every performance metric. This section explains how the benefits should inform controller selection and which extensions require further evidence.

F.1 What nonlinear response shaping adds

The controller offers a local way to change the relationship between occupancy error, its change between measurements and the next frequency correction. Proposition B.2 shows that the normalized min-activation surface strengthens its calibrated linear counterpart for aligned input signs and attenuates its magnitude for opposing signs within the stated central region. A uniform increase in PI gains would scale both cases uniformly. The Fuzzy law therefore supplies a different response shape at calibrated gain coordinates, while retaining a concrete connection to the PI baseline implementation.

This interpretation concerns a correction increment, not a complete closed-loop performance ordering. In particular, aligned signs do not always mean that absolute occupancy error is increasing; an error can cross zero while decreasing in magnitude. The plant, measurement history and accumulated correction determine the resulting trajectory. Equation (71) makes the memory contribution explicit: past nonlinear increments may remain in the accumulator even when later inputs have zero nonlinear residual. This provides

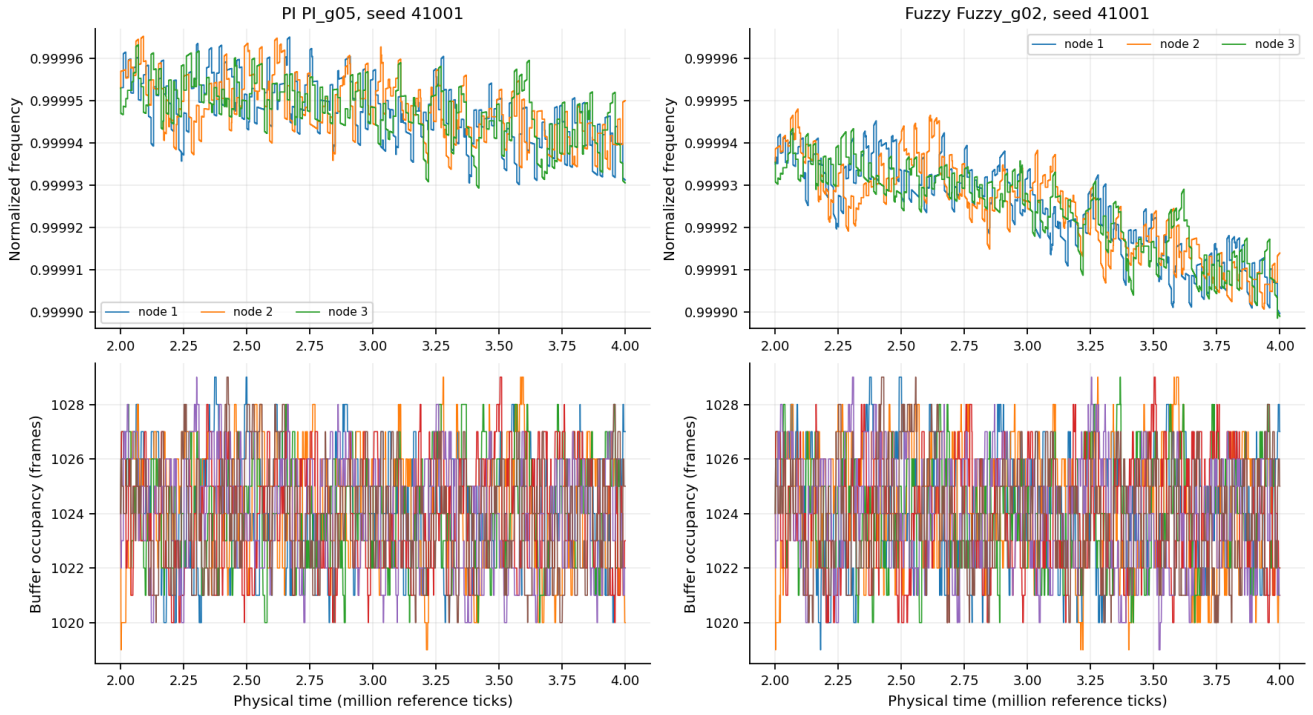


Figure 11: Saved raw traces for the first held-out seed (41001) in the $T_c = 10000$ random experiment. Frequency uses held corrections and shared physical time; buffer occupancy overview samples are for display. Reported moments and buffer extrema instead use complete intervals and events. The paired-seed statistics, rather than this one trace, support the mean performance comparison.

a mechanism to investigate transient shaping and persistent offsets, but a same-input decomposition alone does not establish their causal contribution to differences between independently evolving closed loops.

The comparisons address two questions. The reference proof and startup tests hold the calibrated gain coordinates fixed. They test a prescribed change in the controller law. The equal-budget study instead asks whether each controller family can produce a useful selected design with the same finite search budget. Its lower held-out RMS supports the selected Fuzzy scheme, including its chosen gains, rather than an inference-only effect at identical gains. Keeping these questions separate makes the empirical benefit more meaningful than a comparison against a single arbitrary PI setting.

F.2 Engineering meaning of the buffer-headroom benefit

For a finite receiving buffer, the largest transient deviation matters even when the eventual operating point is admissible. Theorem C.4 compares a certified upper bound for the Fuzzy excursion with an attained PI peak, not just two conservative upper bounds. For its prescribed initialization, the resulting improvement in minimum headroom exceeds 6.0887 fluid frames. Both controllers also satisfy the reference model's positive-frequency, correction and convergence conditions. The benefit thus concerns transient resource use while those conditions remain satisfied.

The matched-gain Callisto tests provide a separate engineering observation. For example, at 1600 ppm, Fuzzy reduces the measured excursion from 94 to 72 frames without correction saturation or gate intervention (Table 10). This is consistent with a useful headroom effect after introducing delayed local execution and integer buffer events, although the mathematical reference certificate does not cover that implementation. At the larger stress scale, fewer saturated updates and smaller excursions remain finite-run findings, not a proof that saturation is harmless or that future overflow is impossible.

These results measure reserve at fixed capacity. They do not establish that a smaller hardware buffer can simply be substituted. Capacity appears in the feedback normalization $\gamma_i = 100/(d_i\beta_{\max})$, where d_i is the number of receiving neighbors. Changing $\beta_{\max}$ or the degree changes the frame-count controller map unless the gains or normalization are adjusted consistently. Any capacity-reduction claim would therefore require a new model-consistent comparison and a fresh safety check. Similarly, a failed sufficient condition is inconclusive: the observed-safe but uncertified reference cases illustrate conservatism, not an enlarged general safe set.

F.3 Selecting a controller for the operating regime

The independent tests demonstrate a practical benefit under the stated selection policy. With identical evaluation budgets, frozen

gains and shared held-out profiles, the random-priority Fuzzy representative reduces RMS inter-node frequency deviation and mean command TV in both original random regimes. The burst-priority representative also substantially reduces IAE in the larger held-out burst. The distinct selected representatives show that the desired task matters: prioritizing frequency deviations under random latency and prioritizing a burst transient need not choose the same parameters. The experiments do not switch between these representatives online.

Several measures are needed to interpret the benefit. Relative frequency agreement concerns differences between nodes; it does not require the common frequency to equal one or to return to its clean counterfactual. The larger common-frequency shift accompanying the selected Fuzzy RMS improvement is therefore an engineering cost to evaluate against the application, not an inconsistency in the RMS measurement. The controller has no absolute-frequency reference loop. Likewise, smaller buffer peaks need not coincide with the lowest RMS, and command TV is an activity metric rather than a measured energy cost. A design choice should specify the acceptable headroom, frequency-shift and correction constraints before selecting among the reported tradeoffs.

The slower-delivery study further shows why timing conditions belong in that choice. Increasing a constant link-latency plateau does not suspend arrivals for the whole plateau: its onset changes in-flight inventory, and the first deficit can form before changed feedback is applied. The ramp profiles allow regulation during the formation of the imbalance, whereas the random profile grids change its temporal structure. Neither a longer hold nor slower variation alone establishes an RMS advantage. These observations motivate matching the controller to a declared disturbance regime; they do not identify an optimal polling period, since polling was not independently swept in the new performance tests.

F.4 Meaning of safety, convergence and logical synchrony

The certificate applies to the synchronous zero-latency fluid model with its compatible targets, initialization and invariant input region. Its sampled recurrence is the exact endpoint map of that continuous zero-order-hold reference. It is not a replacement discrete firing-count plant, nor an exact model of the asynchronous controller scheduling and integer delivery events in Callisto. In particular, the complete 25-rule controller remains executable outside the central input region, while the regional formula and the numerical invariant certificate cannot be applied to arbitrary such trajectories.

The Callisto tests establish capacity compliance, positive frequency and progress over specified finite records. Persistent random input is evaluated by windowed RMS inter-node frequency deviation and occupancy statistics rather than claimed exact convergence to a fixed equilibrium. The startup records also have a shorter declared horizon than the disturbance tests. None of these observations alone supplies an all-time safety or convergence proof for delayed execution. Similarly, the reference frequency-spread result requires a separately justified engineering threshold; logical synchrony does not supply a universal ppm limit.

Logical synchrony is the computational requirement motivating the controller, while physical syntony, buffer centering and positive frequency describe different properties of its implementation. Preserving capacity and frame order, together with continued data progress, supports the normal execution assumed here. Demonstrating functional logical synchrony for an application would additionally require checking the logical frame relationships and payload execution against the corresponding logical model. These experiments do not perform that refinement check or measure application-level useful throughput. A capacity guard that prevents underflow by stopping data progress must also be distinguished from successful unstalled control.

F.5 External validity and next steps

The independent comparison is deliberately bounded: a three-node triangle, a fixed two-dimensional gain grid, fixed Fuzzy input scaling K_e, and one training seed per random regime. The held-out seeds quantify variation within those declared input families; they do not turn the search into global controller optimization or establish performance on larger, heterogeneous or changing networks. The transport model preserves order and permits batch release without a finite serialization rate. Hardware oscillator limits, serialized links, packet loss, node dropout and controller execution cost require separate evaluation. The current observer and supervisory switching mechanisms are disabled, so they contribute no fault-recovery claim.

The most direct theoretical extension is to account explicitly for local sampling, bounded delay and quantization when constructing invariant buffer and input regions. Such a result should preserve absolute-frequency and actuator constraints as well as disagreement bounds. Empirically, broader topologies and delivery models would test whether the observed benefits persist beyond the present triangle. If absolute-rate requirements or sustained saturation are important, a common-frequency constraint or anti-windup mechanism could be studied as a separate controller extension, followed by fresh analysis and independent tests. These steps extend the present mechanism and evidence without changing the scope of its proved or measured benefits.